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Predicting and explaining professionalism issues in medical students with time-to-event models: An exploratory longitudinal learning analytics study

Chang Cai [a], Minyang Chow [a], Ruth Choe [a], Faith Chia Li-Ann [a], Jennifer Anne Cleland [a], Xiuyi Fan [b,*]

[a] Lee Kong Chian School of Medicine, Nanyang Technological University, Singapore, Singapore
[b] Lee Kong Chian School of Medicine, College of Computing & Data Science, Nanyang Technological University, Singapore, Singapore

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ABSTRACT

Medical professionalism links physicians' duties with societal expectations and is fundamental to competent healthcare. Early unprofessional behavior in medical school has been linked to future professionalism issues in clinical practice. However, the longitudinal dynamics and potential predictors of professionalism issues among medical students remain insufficiently explored, with prior research largely focusing on group-level analyses. To address this gap, we developed a time-to-event analytical framework to examine the timing of professionalism issues and factors associated with their occurrence. We analyzed a longitudinal dataset comprising sociodemographic, pre-admission, and in-school performance variables from a single medical school. Using survival analysis, the results suggest that several factors might be associated with the risk of professionalism issues within this institutional context. Specifically, higher pre-admission writing quality scores and academic examination scores were associated with lower risk. In contrast, higher scores in certain pre-admission Multiple Mini Interview (MMI) stations (e.g., Professional role awareness and Professional integrity) were linked to increased risk. Overall, this study proposes a preliminary analytical framework for examining professionalism issues using longitudinal educational data. The framework may support future research on professionalism development in medical training and may provide a methodological approach that could be adapted for use in other educational contexts.

1. Introduction

> **Professionalism** serves as the basis of medicine's relationship with society, and many have termed this relationship a social contract.
>
> —Cruess and Cruess (2008, p. 579)

Medical professionalism underpins modern medicine, serving as a bridge between physicians and society (Cruess et al., 2000). Professionalism is a commitment that privileges patient welfare above self-interest and maintains standards of competence, integrity, and accountability (Project of The ABIM Foundation, 2002). Beyond an ethical ideal, professionalism should also be the moral and practical guide shaping physicians' behavior and influencing the quality of care (Swick, 2000). Professionalism plays a crucial role in fostering patient trust and satisfaction, ensuring safety, and enabling high-quality care (M. D. Brennan & Monson, 2014; Igarashi, 2025). Given its importance, supporting learning to develop professionalism and to act professionally are critical parts of medical education (Cruess et al., 2014; O'sullivan et al., 2012).

Even with increasing focus on supporting the development of professionalism, lapses in professionalism still occur. For example, some medical students display behaviors that raise concerns, typically referred to as unprofessional behavior (Mak-Van Der Vossen et al., 2017). Importantly, research has shown that early unprofessional behaviors in medical school are not isolated incidents but are linked to later professional issues in clinical practice (Krupat et al., 2020; Papadakis et al., 2005). For example, students disciplined for behaviors such as academic dishonesty or interpersonal difficulties are more likely to encounter professional problems during clinical practice (Krupat et al., 2020). Moreover, when graduating medical students fail to meet professional standards, the consequences extend beyond individual outcomes and impose substantial societal costs in the form of wasted

* Corresponding author.
E-mail addresses: CHANG023@e.ntu.edu.sg (C. Cai), minyang.chow@ntu.edu.sg (M. Chow), RuthChoe@ntu.edu.sg (R. Choe), faith_chia@ntu.edu.sg (F.C. Li-Ann), jennifer.cleland@ntu.edu.sg (J.A. Cleland), xyfan@ntu.edu.sg (X. Fan).

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resources and workforce loss (Foo et al., 2018). Given this, early identification of professionalism issues among medical students is crucial.

However, there is a gap in knowledge. Prior research has primarily focused on clarifying what constitutes unprofessional behavior (Igarashi, 2025; Mak-Van Der Vossen et al., 2017; M. Mak-van der Vossen et al., 2020). Questions remain about the extent to which risks can be identified early, and which factors might be linked to unprofessionalism. Prior studies have linked professionalism issues to factors such as reflective ability, sociodemographic context, and academic performance (Hoffman et al., 2016; Papadakis et al., 2005; Paton et al., 2018; Yates & James, 2010). For example, Papadakis et al. (2005) as well as Yates and James (2010) found that poor academic performance in medical school was associated with later unprofessional conduct in clinical practice. However, their (case–control) designs limited the ability to predict who is at risk and when issues may arise. Similarly, previous studies employed traditional statistical analysis that emphasized group-level associations (Krupat et al., 2020; Sahota et al., 2025; Yates & James, 2010), offering limited capacity for individual-level risk analysis and personalized support. Finally, most research has examined professionalism issues in clinical practice (Krupat et al., 2020; Papadakis et al., 2005; Yates & James, 2010), while comparatively little attention has been devoted to their emergence during medical school. These limitations reveal three main gaps: (1) limited methodological approaches for examining longitudinal patterns of professionalism issues, (2) insufficient consideration of the timing of professionalism issues during medical training, and (3) limited analytical frameworks for conducting individual-level analyses of professionalism issues.

Time-to-event analysis offers a methodological approach to address these gaps. Also known as survival analysis, this method has long been used in medicine to model medical prognosis, such as survival with cancer or progression of chronic disease (Altman et al., 1995; Andersen, 1991; Ohno-Machado, 2001). In contrast to conventional regression models, survival analysis explicitly accounts for both whether and when an event occurs (George et al., 2014), while accommodating censored observations. These features are particularly well-suited to educational datasets, where events may occur at different time points and some individuals remain event-free during follow-up (Ameri et al., 2016, pp. 903–912; Wang et al., 2019). In medical education, professionalism issues emerge over time and may occur at different stages of training. Therefore, a time-to-event framework enables the analysis of both the occurrence and timing of such issues. Survival analysis has been used to model student dropout and retention, as well as to identify predictors of academic performance and at-risk students (Ameri et al., 2016, pp. 903–912; Bruinsma & Jansen, 2009; Gury, 2011; Labrador et al., 2019; Martínez-Carrascal et al., 2023; Plank et al., 2005; Wang et al., 2019; Zhuhadar et al., 2019). However, to our knowledge, this is the first time that survival analysis is applied in exploring the timing and associated factors of professionalism issues in medical education.

The primary aim of this study is to apply a time-to-event analytical framework to examine the timing of professionalism issues and factors associated with their occurrence among medical students. Specifically, we address the following research questions:

- **RQ1:** What medical students are at risk of developing professionalism issues, and when are these issues likely to occur?
- **RQ2:** At the class level and the individual level, which variables have the most influence on medical students developing professionalism issues?

This study makes three main contributions. First, we develop a time-to-event analytical framework for examining the occurrence and timing of professionalism issues during medical training. Second, we explore factors associated with professionalism issues using pre-admission, in-school, and sociodemographic data within a single institutional context. Third, we highlight the potential of time-to-event modelling as an analytical approach within learning analytics. The proposed framework may provide a methodological basis for future research on data-informed educational practices.

The remainder of this paper is structured as follows. First, we describe the methodology of the proposed framework. Next, we present the results and discuss their implications for educational practice as well as the study's limitations. Finally, we conclude by summarizing key contributions and outlining directions for future research.

2. Method

2.1. Study design overview

This study developed a three-stage framework integrating prediction, statistical analysis, and individual-level risk analysis. First, a time-dependent Cox (TD-Cox) model (Andersen & Gill, 1982; Cox, 1972) was trained to predict professionalism issues, and key predictors were extracted. Second, statistical analyses were conducted to identify predictors associated with professionalism issues. Third, individual-level risk analysis was performed based on the prediction results. All analyses were conducted in Python using the lifelines package for survival modelling.

2.2. Data collection and description

Data were collected from 1,048 medical students enrolled from 2016 to 2023 at one of Singapore's three medical schools. The study was approved by the Institutional Review Board of our university. Of the 1,048 students, nine individuals with excessive missing values were excluded (two from the 2019 cohort, three from 2020, one from 2021, and three from 2022). After these exclusions, the final analytic sample consisted of 1,039 students.

The variables considered in the study included:

- **Sociodemographic characteristics**: Gender, Age group, and Housing type.[1]
- **Pre-admission performance:** Prior biology knowledge, Biomedical Admissions Test (BMAT)[2] (Cambridge Assessment Admissions Testing, 2003), and Multiple Mini Interviews (MMI)[3] (Eva et al., 2004).
- **In-school performance**: Written examination scores and Objective Structured Clinical Examination (OSCE)[4] (Harden et al., 1975) scores.

The output variable of the study was professionalism issues, defined as a medical student exhibiting unprofessional behavior during medical school. In this study, professionalism issues were evaluated using CHERI (see Table 1), a tool for reporting, and recording data on professionalism issues during medical school. CHERI was developed by our school and was based on the Conscientiousness Index, a validated objective scalar measure of professionalism in undergraduate medical education (McLachlan et al., 2009). Drawing on descriptors of unprofessional behaviors in a previous literature, CHERI classifies student professionalism issues into four domains (Mak-Van Der Vossen et al., 2017): these are

[1] Housing type: students were classified as HDB (public housing) or non-HDB, which served as a proxy socioeconomic indicator.

[2] BMAT: a pre-admission assessment evaluating applicants' critical thinking, scientific knowledge, and writing skills for entry into medical school. It is composed of three sections: Aptitude and Skills, Scientific Knowledge, and Writing.

[3] MMI: a structured series of short stations used before admission to assess applicants' communication, ethical reasoning, and interpersonal skills. Each station evaluates a different competency.

[4] OSCE: a practical exam in which students demonstrate clinical skills. An OSCE is not administered in Year 1; the first OSCE assessment occurs in Year 2.

Table 1
Categories, descriptions, and examples of observable professionalism metrics (CHERI Comments).

Category	Description	Example CHERI Comments
Failure to Engage	Failure to meet assigned responsibilities or tasks	Missing attendance, arriving late at learning sessions, submitting in-course assessments late, not responding to formal communications from the school
Dishonest Behavior	Violations of integrity	Cheating, plagiarism, lying, disregarding rules and regulations
Disrespectful Behavior	Actions that harm peers, faculty, or patients	Disrespectful behavior towards patients, fellow students, colleagues, or teachers
Poor Self-awareness	Difficulties in self-regulation and professional growth	Resisting feedback, blaming others, failing to recognize limitations

described below and summarized in Table 1.

- **Failure to engage:** refers to neglect of duties, including lateness, absenteeism, missed deadlines, low motivation, and poor teamwork (Keshmiri & Raadabadi, 2023; Mak-; Papadakis et al., 1999; Song & Willy, 2024; Van Der Vossen et al., 2017; Ziring et al., 2015).
- **Dishonest behavior:** encompasses integrity violations such as cheating, plagiarism, and disregard for regulations (Ainsworth & Szauter, 2006; da Rosa et al., 2024; Keshmiri, 2025; Keshmiri & Raadabadi, 2023; Mak-Van Der Vossen et al., 2017; Yates, 2014; Ziring et al., 2015).
- **Disrespectful behavior:** involves inappropriate interactions with peers, staff, or patients, including rudeness, lack of empathy, or unprofessional communication (Keshmiri, 2025; Mak-; Song & Willy, 2024; Van Der Vossen et al., 2017; Yadav et al., 2019; Ziring et al., 2015).
- **Poor self-awareness:** reflects inadequate self-regulation, often manifested as resistance to feedback or refusal to change (Mak-Van Der Vossen et al., 2017; Papadakis et al., 1999; Ziring et al., 2015).

Students with one or more documented CHERI comments in any academic year were coded as 1 (issue present), whereas all other students were coded as 0 (no issue). Only the first occurrence of unprofessional behavior for each student was considered, as the earliest identification is critical for timely intervention. In addition, the first recognized instance of unprofessional behavior typically triggers remediation or disciplinary action, which may subsequently modify the student's behavior and thereby confound longitudinal analyses. Among all the students, 418 (40.23%) experienced professionalism issues, while 621 (59.77%) did not. Table 2 presents the distribution of events across the five study years.

2.3. Data processing

Missing numerical values were imputed using either means or rounded means (Donders et al., 2006) for the TD-Cox modelling. For variables originally recorded as integers, the rounded mean was used.

Table 2
Summary of the final sample characteristics (N = 1039).

Category	Count	Percentage
No professionalism issue	621	59.77%
With professionalism issue	418	40.23%
Year 1 event	113	10.88%
Year 2 event	92	8.85%
Year 3 event	119	11.45%
Year 4 event	57	5.49%
Year 5 event	37	3.36%

For continuous variables, the arithmetic mean was applied without rounding. For both written and OSCE examinations, summary statistics, such as average, maximum, minimum, and change scores (Year_t Score − Year_{t-1} Score and $|\text{Year}_t$ Score − Year_{t-1} Score$|$), were computed. In addition, both the raw difference and the absolute difference between written and OSCE scores were calculated. Then, Spearman's rank correlation method (Spearman, 1961) was applied as a preprocessing step for numeric feature selection. For any pair of variables with correlations of $|\rho| \geq 0.7$, one variable was removed and the other retained. Appendix Table A1 summarizes all variable pairs with strong correlations ($|\rho| \geq 0.70$), including their correlation coefficients (ρ) and p-values. Appendix Figures A1 and A2 present the full Spearman correlation matrices. Appendix Table A2 presents detailed information about the variables included in the analysis.

Categorical variables were encoded using one-hot encoding. Numerical variables were standardized using Z-score, whereby each variable was centered by subtracting the mean and scaled by dividing the standard deviation. To identify key predictors and avoid data leakage, only variables available prior to the occurrence of professionalism issues were used. For example, if a student developed issues in the third year, only sociodemographic variables, pre-admission scores, and performance on OSCE and written examinations from the first two years were considered.

2.4. Prediction modelling (time-to-event analyses)

This study uses longitudinal data in which each student is followed over multiple years, and the timing of professionalism issues differs across individuals. Some students experience an event early, some later, and others remain event-free and thus censored. Because the outcome of interest was the timing of the first occurrence of professionalism issues, a time-to-event framework was used. This approach allows the analysis to account for both the timing of events and censoring among students who remained event-free by the end of follow-up.

In our time-to-event analyses, we first defined key terms used throughout this paper:

- **Event**: In this study, the event was the first occurrence of documented unprofessional behaviors of medical students. Event = 1 indicates that a student exhibited unprofessional behavior, whereas Event = 0 indicates no observed event (censored).
- **Hazard:** The hazard represents the instantaneous rate at which a student may experience a professionalism issue at time t. It reflects how fast medical students exhibit unprofessional behaviors.
- **Cut-off time point**: The cut-off time point corresponded to the maximum observation period available for each cohort, beyond which no further follow-up data were collected. Specifically, the cut-off time point was set at Year 5 for students enrolled in 2016–2019, Year 4 for 2020, Year 3 for 2021, Year 2 for 2022, and Year 1 for 2023.
- **Censoring**: If a student did not exhibit professionalism issues by a cut-off time point, the observation was treated as censored. Survival analysis can effectively handle censored data (Ameri et al., 2016, pp. 903–912). As shown in Fig. 1, students B, C, D, E, and F (non--censored) exhibited professionalism issues before the cut-off time point, whereas Student A (censored) did not exhibit such issues before the cut-off time point.
- **Time intervals**: The five intervals considered were the end of Year 1, Year 2, Year 3, Year 4, and Year 5.
- **Linear predictor** ($\eta_i(t)$): For each student i at time t, the linear predictor is defined as $\eta_i(t) = \sum_{j=1}^{p} \beta_j x_{ij}(t)$, where β_j are the regression coefficients estimated by the TD-Cox model and $x_{ij}(t)$ are the variable values for student i at time t.
- **Hazard ratio (HR)**: HR refers to the relative instantaneous risk of professionalism issues associated with changes in variable values.

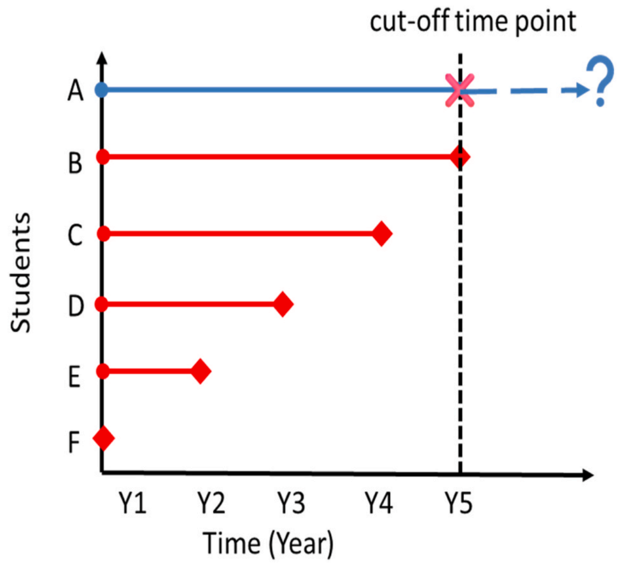

Fig. 1. An illustration to demonstrate the student's professionalism issues. In this example, Students B, C, D, E, and F exhibited professionalism issues in Years 5, 4, 3, 2, and 1, respectively. Student A did not exhibit professionalism issues and was therefore censored.

HR > 1 indicates increased risk, whereas HR < 1 indicates a protective effect.

TD-Cox Model. Because some variables (e.g., written and OSCE scores) varied over time, we used a TD-Cox model. The TD-Cox model extends the classical Cox proportional hazards model to simultaneously handle both static and time-dependent covariates (Ameri et al., 2016, pp. 903–912). Pre-admission and sociodemographic variables were treated as statics variables, whereas in-school performance measures were updated across academic years as time-varying variables.

For TD-Cox analysis, the dataset was restructured using the counting process approach, which accommodates time-varying covariates (Andersen & Gill, 1982). As shown in Fig. 2, the data were transformed from a record-per-student format into a record-per-interval (long) format. The long-format dataset information is provided in Appendix Table B1.

We then used the TD-Cox model to predict students at risk of developing professionalism issues. We stratified the TD-Cox model by academic year so that the baseline hazard was allowed to vary across Years 1–5. The dataset was partitioned into training (80%) and test (20%) sets, with the split performed at the student level. Within the training set, five-fold cross-validation was conducted to assess stability. The predictive model was constructed using all variables (see Appendix Table A2). Based on the survival function estimated from the TD-Cox model, we computed the predicted event probability for the subsequent time interval, denoted as p_{next}. p_{next} represented the conditional probability of experiencing the event in the interval $(t-1, t]$, given that the student had survived without issues up to time $t-1$. To transform the probability into a binary classification outcome, we compared p_{next} with a threshold defined as the observed event rate in the training set. Students with $p_{\text{next}} \geq$ threshold were classified as at risk of developing professionalism issues. To evaluate the predictive performance of the proposed model, we adopted commonly used metrics from survival and classification analysis:

- **Concordance index (C-index)** (Harrell Jr et al., 1996): A widely used performance measure for survival models, the C-index evaluates how accurately the model ranks predicted survival times (Steck et al., 2007). It is a generalization of the area under the ROC curve (AUC) that accounts for censored data. A C-index of 1.0 indicates perfect prediction, while a value of 0.5 suggests a random prediction.
- **Brier score** (Glenn, 1950): The Brier score quantifies both discrimination and calibration of a survival model based on its estimated survival functions. It extends the mean squared error to right-censored data, with lower values indicating better overall predictive accuracy.
- **Accuracy**: A metric that measures the proportion of correctly classified instances.
- **F1-score (weighted)**: The mean of precision and recall.

Key Predictors Identification from the Predicting Model. To identify key predictors, we ranked variables according to their HRs and *p*-values. HR represents the ratio of instantaneous risk associated with a one-unit change in a variable. We also calculated the 95% confidence intervals for each HR. *p*-values less than 0.05 were considered statistically significant.

2.5. Statistical analyses

Univariable Analyses (Kaplan–Meier Estimator (Kaplan & Meier, 1958)**).** To examine the predictive power of individual variables on the timing and likelihood of professionalism issues, we performed survival analyses using the Kaplan–Meier estimator. This method shows how the number of students exhibiting unprofessional behaviors changes over time. Specifically, the Kaplan–Meier estimator accounts for (a) the time to event for each student, (b) the censoring status, and (c) the risk set at each time point. Kaplan–Meier curves therefore show the probability of remaining professional from Years 1–5. Kaplan–Meier curves were used as an exploratory univariable visualization to compare survival patterns across groups. For visualization, continuous variables were dichotomized into high- and low-score groups to generate Kaplan–Meier curves. Predefined rules were applied for dichotomization: for most variables, the median was used as the cut-off. For BMAT_S3B, where most students achieved the maximum score (5), the high group was defined as score 5 and the low group as scores below 5. Kaplan–Meier survival curves were generated for each grouping, and differences between groups were

Student ID	First_Issue_Year	Event	Features
ID_1	N	0	...
ID_2	2	1	...
ID_3	1	1	...

Example of survival data.

Student ID	Start	Stop	Year_t	Event	Features
ID_1	0	1	1	0	...
ID_1	1	2	2	0	...
ID_1	2	3	3	0	...
ID_1	3	4	4	0	...
ID_1	4	5	5	0	...
ID_2	0	1	1	0	...
ID_2	1	2	2	1	...
ID_3	0	1	1	1	...

Example of survival data after counting process based reformatting.

Fig. 2. An example of reformatting the data using the counting process.

assessed using the log-rank test (Bland & Altman, 2004). For variables identified as significant by the Kaplan–Meier method, we further examined differences by comparing the top 10% and bottom 10% of students.

Multivariable Analyses (TD-Cox Model). While the Kaplan–Meier estimator provided univariable insights into how predictors influenced the timing and likelihood of professionalism issues, it did not account for the joint effects of multiple predictors. Therefore, we fitted a TD-Cox regression model to examine which factors were associated with increased or decreased risk of developing professionalism issues over time. This model is appropriate for longitudinal data as it accommodates static and time-varying variables, allowing the hazard to be updated over time. This approach estimated HR, indicating how many times more (or less) likely a participant was to develop professionalism issues. Variables identified from the Kaplan–Meier analyses and prediction model were further examined in a multivariable TD-Cox model to evaluate their joint effects on professionalism issues. This approach integrates predictors identified based on both statistical association and predictive relevance. The model was stratified by academic year to allow year-specific baseline hazards. Regression coefficients and corresponding HRs were estimated. 95% confidence intervals were also calculated for each HR. The significance level was set at 0.05.

Exploratory Analysis by Professionalism Issues Severity. In addition to the binary classification of professionalism lapses, we conducted exploratory analyses stratified by professionalism issues severity (minor vs. severe issues). Based on our CHERI framework (see Table 1), "Failure to Engage" was classified as a *minor* issue, while "Dishonest Behavior", "Disrespectful Behavior", and "Poor Self-awareness" were categorized as *severe* issues. Group comparisons were conducted according to variable types. Categorical variables were analyzed using the Chi-square test, with Fisher's exact test used when more than 20% of cells had expected counts below five (Nowacki, 2017). For continuous variables, normality was assessed with the Shapiro–Wilk test. For normally distributed data, variance was tested with Levene's test, followed by a *t*-test or Welch's *t*-test as appropriate. Non-normally distributed variables were analyzed using the Mann–Whitney *U* test (Du Prel et al., 2010).

2.6. Individual-level risk analysis

In line with the principles of precision education, where interventions are personalized (Coates, 2025; Rajasekaran, 2024; Triola & Burk-Rafel, 2023), we identified individualized factors and generated personalized probability trajectories.

Specifically, to identify key factors for each student, we quantified the contribution of each variable to an individual's predicted risk. Each variable's contribution to the individual risk was derived by multiplying its estimated regression coefficient with the student's corresponding variable value at the latest observed interval. Positive contributions indicated increased risk, whereas negative values represented protective effects. Subsequently, the key individual predictors were identified by ranking the absolute magnitudes of their contributions. To capture temporal dynamics, we further summarized each student's predicted risk trajectory across academic years.

3. Results

3.1. Overview of results

The predictive model performed well overall. Both pre-admission and in-school variables contributed to the prediction of professionalism issues. Table 3 summarizes predictors that were significant or marginally significant ($p < 0.06$) in both Kaplan–Meier and TD-Cox analyses. Individual-level risk analysis further illustrated how the relative importance of predictors varied across students. The following sections present these findings in detail.

Table 3
Summary of predictors of professionalism issues.

Category	Predictors	Direction of Association	Analysis for Survival Timing
Pre-admission performance	Written content quality (BMAT_S3A)	Higher scores → lower risk [a]	Associated with longer survival; slower onset of professionalism issues
Pre-admission performance	Professional role awareness (MMI_S1), Professional integrity (MMI_S6), Written English quality (BMAT_S3B)	Higher scores → higher risk [a]	Associated with shorter survival; faster onset of professionalism issues
In-school performance	Academic performance (Written_mean)	Higher scores → lower risk [b]	Associated with longer survival; slower onset of professionalism issues
Sociodemographic characteristics	Gender, Age group, Housing type	Not significant	No meaningful

Note. Variables shown were significant or marginally significant ($p < 0.06$) in either Kaplan–Meier or TD–Cox analyses. [a] Significant in both Kaplan–Meier and TD-Cox analyses. [b] Significant in Kaplan–Meier and marginally significant in TD-Cox analyses.

3.2. Prediction performance and key predictors

Table 4 summarizes the predictive performance across test sets by year. Overall, the predictive framework achieved satisfactory performance, with accuracy consistently around 0.8 and F1-scores ranging from 0.7505 to 0.8624. For the C-index, the best results were obtained for Years 1 and 3, which showed the highest scores (0.7543 and 0.8050, respectively). However, the C-index for Years 4 and 5 was comparatively lower (C-index = 0.4121 and 0.4812), likely due to smaller sample sizes ($n = 92$ and $n = 64$ respectively). Brier scores remained low across all years (0.0880–0.1564), suggesting a good overall calibration of the model.

Fig. 3 shows the fifteen variables with the highest predictive power for predicting professionalism issues. Professional integrity (MMI_S6) and Writing content quality (BMAT_S3A) were statistically significant ($p < 0.05$), with hazard ratios of 1.1030 and 0.8970, respectively. Specifically, each one-unit increase in Professional integrity (MMI_S6) corresponded to an approximately 10.30% increase in risk, whereas each one-unit increase in Writing content quality (BMAT_S3A) corresponded to an approximately 10.30% decrease in risk. The complete results are available in Appendix Table B2.

3.3. Effects of predictors on professionalism issues

3.3.1. Univariable results based on Kaplan–Meier curves

Kaplan–Meier curves revealed significant differences in survival probabilities across several pre-admission and in-school variables. However, no significant differences were observed for sociodemographic variables (including Gender, Age Group, and Housing Type) in relation to professionalism issues.

For pre-admission variables (see Fig. 4), students with higher BMAT

Table 4
Performance of TD-Cox model on the test dataset.

Year	C-index	Brier	Accuracy	F1	Sample Size
1	0.7543	0.0880	0.8654	0.8624	208
2	0.6375	0.1274	0.8166	0.7941	169
3	0.8050	0.1564	0.8000	0.7973	130
4	0.4121	0.1342	0.7717	0.7505	92
5	0.4812	0.0983	0.8281	0.8331	64

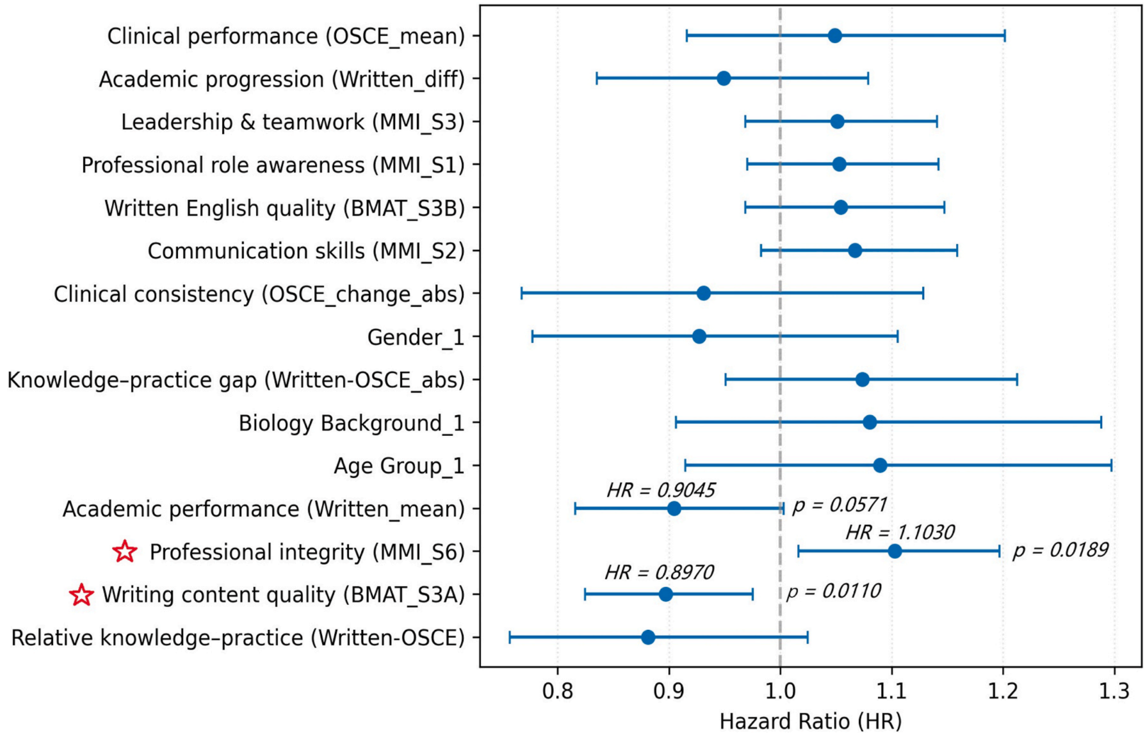

Fig. 3. *Forest plot displaying HR with 95% CI from the TD-Cox model. HR > 1 indicates increased risk, HR < 1 decreased risk, and estimates crossing 1 are non-significant. The red stars denote statistically significant predictors (p < 0.05).*

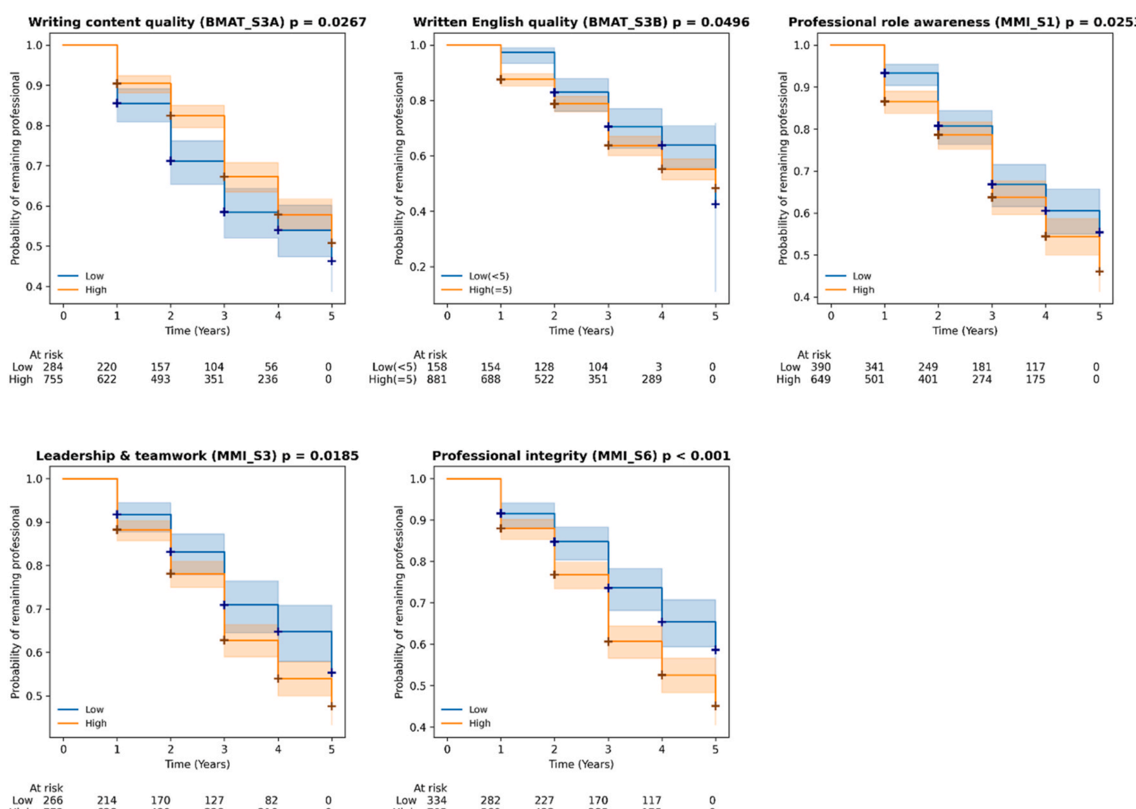

Fig. 4. Kaplan–Meier curves compared students with low and high pre-admission scores. The y-axis represents the probability of remaining professional (survival probability), where higher values indicate a lower risk of professionalism issues. Cross marks (+) indicate censored data (students who graduated or left the study without a recorded professionalism issue).

writing content quality (BMAT_S3A) consistently showed higher survival probabilities. Their Kaplan–Meier curves showed more gradual declines compared with lower-scoring students, reflecting a slower progression toward professionalism issues. In contrast, higher scores in BMAT_3B (Written English quality) and MMI stations (Professional role awareness (MMI_S1), Leadership & teamwork (MMI_S3), Professional integrity (MMI_S6)) were associated with lower survival probabilities. The Kaplan–Meier curves for these MMI variables showed earlier and steeper declines, reflecting that professionalism lapses occurred sooner in this group. Other variables were not significantly associated.

For in-school performance variables (see Fig. 5), Kaplan–Meier analyses showed lower survival probabilities among students with weaker written performance (Academic performance (Written_mean)) and greater score fluctuations (Academic fluctuation (Written_change_abs)). The steeper decline of these curves reflected that professionalism lapses occurred faster in these groups. Additionally, higher OSCE scores (Clinical performance (OSCE_mean)) and larger discrepancies between written and clinical scores (Written–OSCE_abs) were both associated with lower survival probabilities. In both cases, the steeper decline of the Kaplan–Meier curves reflect an earlier occurrence of professionalism lapses. In contrast, students with higher written scores and relatively lower OSCE scores (Relative knowledge–practice (Written-OSCE) > 0) showed higher survival probabilities, with flatter curves and a later occurrence of professionalism issues. Other variables were not significantly associated.

To further evaluate the observed associations, Kaplan–Meier curves were generated for students in the top 10% and bottom 10% of each predictor (see Appendix Figures C1 and C2). For pre-admission predictors, Writing content quality (BMAT_S3A), Professional role awareness (MMI_S1), and Professional integrity (MMI_S6) showed significantly different between groups, consistent with earlier findings. Specifically, students in the top 10% of BMAT writing content quality (BMAT_S3A) consistently showed higher survival probabilities, with flatter curves reflecting a lower likelihood and delayed onset of professionalism issues. In contrast, students in the top 10% of Professional role awareness (MMI_S1) and Professional integrity (MMI_S6) displayed lower survival probabilities, and the steeper decline of their curves indicates a faster onset of professionalism lapses. For in-school predictors, students in the top 10% of Academic performance (Written_mean) and Relative knowledge–practice (Written-OSCE) showed higher survival probabilities, accompanied by flatter curves consistent with later onset. In contrast, the top 10% of Clinical performance (OSCE_mean) and those with the larger Knowledge–practice gaps (Written–OSCE_abs) displayed lower survival probabilities with steeper declines, reflecting earlier professionalism issues. Other measures (Written English quality (BMAT_S3B), Leadership & teamwork (MMI_S3), and Academic fluctuation (Written_change_abs)) were not statistically significant in this subgroup comparison.

3.3.2. Multivariable results based on TD-Cox regression analysis

TD-Cox regression identified several significant predictors of professionalism issues. As shown in Table 5 and Fig. 6, higher BMAT writing content quality (Writing content quality (BMAT_S3A)) was protective, with each one-unit increase associated with an 18.22% reduction in hazard, indicating a slower progression to the professionalism issue (HR $= 0.8178$, 95% CI [0.7374, 0.9069], $p < 0.001$). In contrast, higher scores in Professional integrity (MMI_S6) (HR $= 1.2188$, 95% CI [1.1041, 1.3454], $p < 0.001$), Written English quality (BMAT_S3B) (HR $= 1.1581$, 95% CI [1.0381, 1.2920], $p = 0.0085$), and Professional role awareness (MMI_S1) (HR $= 1.1197$, 95% CI [1.0164, 1.2335], $p = 0.0221$) were associated with increased hazard, corresponding to faster onset of unprofessional behaviors. Each one-unit increase in Professional integrity (MMI_S6) corresponded to an approximately 21.88% higher hazard, Written English quality (BMAT_S3B) to a 15.81% higher hazard, and Professional role awareness (MMI_S1) to an 11.97% higher hazard. Academic performance (Written_mean) showed marginal significance $p = 0.0547$.

These patterns were consistent with the Kaplan–Meier curves. Predictors associated with lower hazard ratios (e.g., BMAT_S3A) corresponded to slower progression to unprofessional behavior and longer

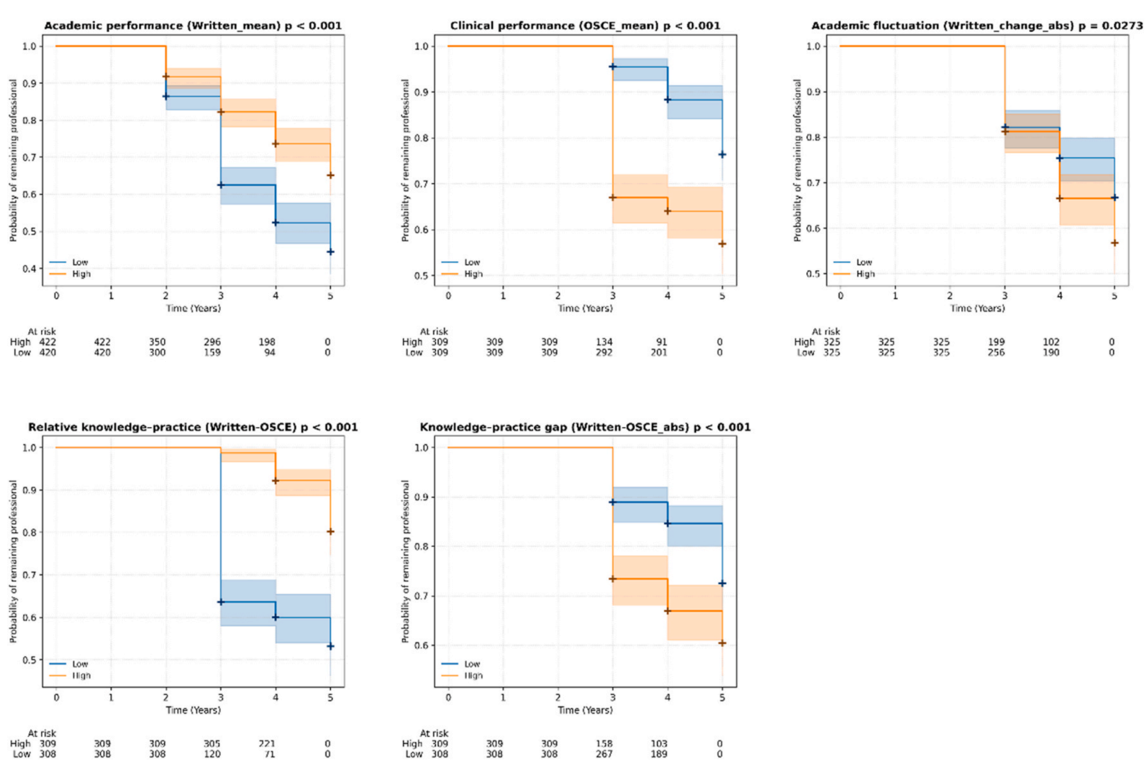

Fig. 5. Kaplan–Meier curves compared students with low and high in-school scores. The y-axis represents the probability of remaining professional (survival probability), where higher values indicate a lower risk of professionalism issues. Cross marks (+) indicate censored data (students who graduated or left the study without a recorded professionalism issue).

Table 5
TD-Cox regression results for predictors of professionalism issues.

Variable	p	95% CI for HR	HR	Coef
Professional integrity (MMI_S6)	**< 0.001**	**[1.1041, 1.3454]**	1.2188	0.1979
Relative knowledge–practice (Written-OSCE)	0.0998	[0.6348, 1.0404]	0.8127	−0.2074
Writing content quality (BMAT_S3A)	**< 0.001**	**[0.7374, 0.9069]**	0.8178	−0.2012
Written English quality (BMAT_S3B)	**0.0085**	**[1.0381, 1.2920]**	1.1581	0.1468
Academic performance (Written_mean)	0.0547	[0.7389, 1.0030]	0.8609	−0.1498
Knowledge–practice gap (Written-OSCE_abs)	0.0871	[0.9819, 1.3091]	1.1338	0.1256
Professional role awareness (MMI_S1)	**0.0221**	**[1.0164, 1.2335]**	1.1197	0.1131
Academic fluctuation (Written_change_abs)	0.1272	[0.7698, 1.0332]	0.8918	−0.1145
Clinical performance (OSCE_mean)	0.4234	[0.8827, 1.3457]	1.0899	0.0861
Leadership & teamwork (MMI_S3)	0.1655	[0.9718, 1.1817]	1.0716	0.0692

Note. HR = hazard ratio; CI = confidence interval; Coef = regression coefficient (ln (HR)). Statistical significance is defined as $p < 0.05$, which corresponds to 95% confidence intervals for HR that do not include 1.

survival, whereas predictors with higher hazard ratios (e.g., MMI sections) were associated with a more rapid emergence of professionalism issues.

3.3.3. *Analysis of professionalism issues severity*

Of the 418 students who had professionalism issues, most were minor issues such as “Failure to Engage” (301, 72.01%), whereas a smaller subset involved severe unprofessional behaviors (“Dishonest Behavior”, “Disrespectful Behavior”, and “Poor Self-awareness”) (117, 27.99%).

Students with severe professionalism issues showed significantly greater clinical fluctuation (OSCE_change_abs; $p = 0.046$) (Appendix Figure D1). By contrast, sociodemographic variables showed no significant or marginal differences between the two groups. Detailed statistical results are provided in Appendix Figure D1 and Appendix Table D1.

3.4. *Examples of individual-level risk analysis results*

We illustrated the individual-level risk analysis results with two case examples: Student N, who did not develop professionalism issues over five years, and Student P, who experienced an issue in Year 3 (Appendix Figure E1).

For each student, individualized analytics summarize the predictors most influential to the learner's risk profile, the cumulative contribution of these predictors to the hazard estimate, the individualized survival trajectory, and the predicted probability of experiencing a professionalism issue over time. The complete visualizations are provided in Appendix Figure E1. These examples illustrate how individual-level analytics can reveal distinct risk patterns in students with and without professionalism issues.

4. Discussion

Medical professionalism is a cornerstone of medical education and practice (Cruess et al., 2000; O'sullivan et al., 2012), and unprofessional behavior during medical school predicts future professionalism lapses (Krupat et al., 2020; Papadakis et al., 2005). Managing unprofessional behavior involves six stages: definition, prevention, detection, evaluation, correction, and follow-up (Igarashi, 2025). However, most previous studies have primarily focused on the defining stage, with comparatively limited attention to longitudinal analysis. To address these gaps, this study introduces a time-to-event analysis framework. This framework treats professionalism as a dynamic developmental process rather than a static outcome. Using a TD-Cox model, we examined longitudinal data to provide preliminary estimates of both the likelihood and timing of professionalism issues. We further identified key factors at both the class and individual levels. Taken together, this study highlights the methodological value of longitudinal learning analytics in examining medical professionalism development.

4.1. *Main findings*

In our setting, in-school performance (Written_mean) and pre-admission indicators (MMI station scores; BMAT subscores) showed potential associations with the risk of developing professionalism issues. No significant associations were observed between sociodemographic

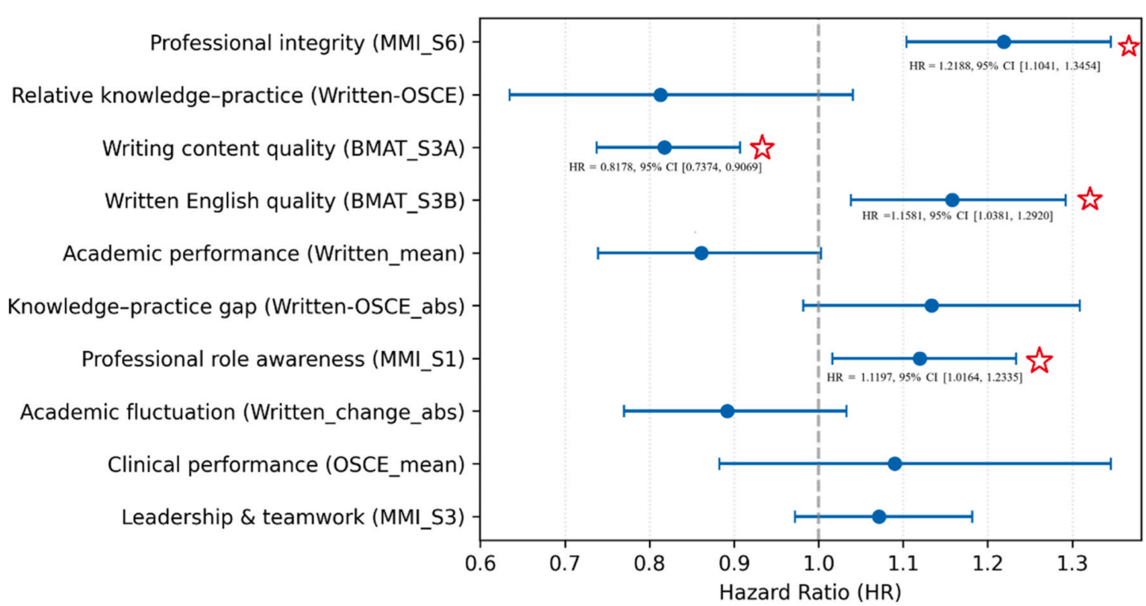

Fig. 6. *Forest plot displays HR with 95% CI from the TD-Cox regression analysis. An HR greater than 1 indicates increased risk; an HR less than 1 indicates decreased risk, while HR estimates crossing 1 suggest non-significant effects. The red stars denote statistically significant predictors ($p < 0.05$).*

variables and professionalism issues. Firstly, consistent with prior work, lower written performance during medical school was associated with a trend toward increased risk of professionalism issues (Cuddy et al., 2017; Krupat et al., 2020; Papadakis et al., 2005), although this association was marginal in the present dataset.

Secondly, a **notable and unexpected** finding was that *higher pre-admission scores on two MMI domains (Professional role awareness, MMI_S1; Professional integrity, MMI_S6) were associated with increased risk of professionalism issues.* Students with higher scores in these areas showed faster progression toward professionalism issues. Across most academic years from 2016 to 2023, MMI_SI and MMI_S6 showed comparable or higher means/medians in the issue group (see Appendix Tables F3–F4 and Figures F2–F3). We emphasize that these findings represent associations rather than causal relationships. A possible explanation for this is that the structured format of MMIs forces students to "perform" rather than respond naturally (Ali et al., 2019, pp. 1031–1038). This implies that students may display appropriate behaviors without fully internalizing professional norms and values. This is consistent with concerns raised by Monrouxe et al. (2011) regarding students who "act" professionally without truly "being" professionally. Another possible reason may be the difference between the interview setting and real practice, such as time pressure, case complexity, and unpredictability. Performance elicited under predictable, highly structured, and observed interview conditions may not generalize to the complex, time-pressure, and frequently unobserved circumstances of routine clinical practice. Finally, MMIs may not capture a candidate's capacity for long-term growth or adaptability in clinical settings (Roberts et al., 2025). Accordingly, scores from cue-rich interview settings may not fully generalize to contexts in which professionalism is enacted over time and under uncertainty. This may indicate a potential professional authenticity gap, with some misalignment between behaviors sampled in brief, examiner-active selection tasks and those required in sustained, examiner-passive work. Importantly, the present findings should be viewed as exploratory and hypothesis-generating, and further research across multiple institutions will be required to determine whether similar patterns are observed in other contexts.

Thirdly, *lower BMAT Written content quality (BMAT_S3A) was associated with greater risk and a faster progression toward unprofessional behavior, whereas higher Written English quality (BMAT_S3B) showed the opposite pattern.* However, as BMAT was used by only a small number of medical schools and has been recently discontinued, we have also repeated our analysis without BMAT with results presented in Appendix F. Overall, the findings remained broadly consistent. When BMAT was removed from the model, Academic performance (Written_mean) was associated with lower risk, whereas Professional integrity (MMI_S6) and Professional role awareness (MMI_S1) were associated with higher risk.

Fourthly, in contrast to prior research, our findings showed no significant association between students' sociodemographic background and professionalism issues. Previous studies have reported that gender was related to professionalism-related disciplinary outcomes (Paton et al., 2018). Similarly, Yates and James (2010) reported that students involved in professionalism-related disciplinary cases were disproportionately male and from lower social class backgrounds. This discrepancy may reflect contextual differences between study localities. In Singapore, government support reduces socioeconomic disparities in access to higher education (Ministry of Education [MOE], 2025). Future research should investigate how sociodemographic factors interact with cultural and institutional contexts to influence the occurrence of professionalism lapses.

Finally, although the predictive performance of the TD-Cox model was generally satisfactory, we observed variability across academic years, particularly in Year 4 and Year 5. This variability may reflect characteristics of the data, given that similar patterns were observed across one robustness check and two sensitivity analyses (Appendix Tables F1, F5, and F7). The stability of Cox model estimates depends on having sufficient events within each time interval (Peduzzi et al., 1995). In the present study, later academic years had substantially smaller sample sizes and fewer observed events. This may reduce statistical power and lead to unstable variance estimates. As a result, the reliability of regression coefficients may be affected (Peduzzi et al., 1995), which may compromise predictive performance. Several potential sources of bias should also be considered. Professionalism issues were derived from institutional records, which may be influenced by variability in reporting practices across cohorts or training stages. Academic performance indicators may also reflect differences in assessment design or curriculum structure over time. These sources of measurement and reporting variability may have a greater impact on smaller samples. Furthermore, one key predictor (academic performance) demonstrated only marginal statistical significance and should therefore be interpreted cautiously. Although two sensitivity analyses yielded broadly consistent results (Appendix Tables F2 and F6; Figures F1 and F4), the findings should be regarded as preliminary and require validation in larger and independent datasets.

4.2. Educational implications

This study has both theoretical and methodological implications for education. Theoretically, the findings contribute to ongoing discussions of precision education in the context of medical professionalism development. By modelling professionalism issues as time-dependent events rather than static outcomes, this work illustrates how survival analytics can be used to analyze the timing of professionalism issues. Moreover, some observed patterns may provide exploratory insights for understanding professionalism issues. For example, the gap between strong MMI performance and later professionalism issues may indicate differences between performance in structured assessments and real contexts, which is consistent with limitations of situated learning. However, these interpretations are exploratory and intended to generate hypotheses for future research on the causal mechanisms underlying these associations.

Methodologically, the study offers a time-to-event analytical framework to analyze longitudinal educational data at individual learner level. This perspective aligns with developments in learning analytics and precision education, which advocates for moving beyond "one-size-fits-all" instruction toward personalized support (Coates, 2025; Rajasekaran, 2024; Triola & Burk-Rafel, 2023). This analytical approach may also be applicable to other educational outcomes where both occurrence and timing are important, such as dropout or academic risk monitoring. While the findings are institution-specific, the framework offers a preliminary tool for examining longitudinal educational patterns and may inform future learning analytics research.

4.3. Limitations

This study has several limitations. First, it was conducted within a single medical school using institution-specific definitions of professionalism issues. Interpretations of professionalism are shaped by cultural values and norms, which vary across countries or cultures (Helmich et al., 2017). Therefore, the findings should be interpreted as context-dependent and exploratory, and require validation in multi-institutional datasets. Nevertheless, the methodological framework is transferable. Institutions could retrain the model using local data to derive contextually relevant predictors and risk trajectories.

Second, professionalism issues were derived from institutional records and may be subject to reporting and measurement bias. Variability in documentation practices across cohorts may introduce bias, which might affect model performance, particularly in smaller samples.

Third, performance variability across academic years, especially in later cohorts, reflects limited sample sizes and fewer observed events. In addition, some predictors showed only marginal statistical significance and should be interpreted cautiously, despite broadly consistent findings across robustness and sensitivity analyses.

Fourth, the sample was imbalanced across different severity levels of professionalism issues, which limited our analyses to exploratory group comparisons. This study aimed primarily to build a framework to explore who may be at risk of developing professionalism issues and when such issues might arise. Future research with larger datasets could employ more fine-grained categorizations to better capture the complexity of professionalism lapses.

Finally, the current framework focuses on performance among medical students and does not extend to clinical practice or postgraduate settings. Studies that follow students into residency or early career stages are needed to determine whether the identified risk factors persist beyond medical school.

5. Conclusion

This study highlights the methodological value of using longitudinal data to examine professionalism issues in medical education. Our framework provides a preliminary analytical approach for examining the occurrence and timing of professionalism issues. The findings suggest that in-school and pre-admission variables may be associated with professionalism issues. However, this study is exploratory, and future studies should validate this framework and findings across multiple institutions to enhance generalizability. Future research can also include expert validation of the identified factors to ensure their practical relevance. In addition, future research can explore more fine-grained approaches to definite professionalism issues. For example, future studies could construct unprofessional behavior profiles based on different patterns of issues (M. C. Mak-van der Vossen et al., 2016) and predict their occurrence. Moreover, longer term studies are needed to link professionalism in medical school with later clinical practice. Such work could integrate in-school assessments with post-graduation outcomes, including patient evaluations and qualitative feedback.

Overall, this study contributes to learning analytics and medical education by introducing a time-to-event analytical approach for examining professionalism issues using longitudinal data. This approach may inform future research on professionalism development and other longitudinal educational processes.

CRediT authorship contribution statement

Chang Cai: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Minyang Chow:** Writing – review & editing, Data curation, Conceptualization. **Ruth Choe:** Writing – review & editing, Data curation, Conceptualization. **Faith Chia Li-Ann:** Writing – review & editing, Data curation, Conceptualization. **Jennifer Anne Cleland:** Writing – review & editing, Supervision. **Xiuyi Fan:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Availability of data and materials

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1 shows the pairs of variables with strong Spearman correlations ($|\rho| > 0.70$) and their *p* values. As shown in Figure A1, ten variables exhibited high collinearity ($|\rho| \geq 0.7$). Thus, we removed the following variables: Written_prev, OSCE_prev, Written_prev, Written_max_sofar, Written_min_sofar, OSCE_max_sofar and OSCE_min_sofar. After removal, a subsequent Spearman correlation check confirmed that the remaining pair of variables did not exceed the $|\rho| \geq 0.7$ threshold (see Figure A2). Table A2 shows the variables we used in this study.

Table A1
Variable pairs with strong Spearman's correlations ($|\rho| > 0.70$)

Variable 1	Variable 2	ρ (rho)	*p*_value
OSCE_prev	OSCE_min_sofar	0.9418	<0.001
Written_mean	Written_min_sofar	0.9034	<0.001
Written_prev	Written_mean	0.8913	<0.001
Written_mean	Written_max_sofar	0.8862	<0.001
Written_prev	Written_max_sofar	0.8536	<0.001
OSCE_prev	OSCE_mean	0.8430	<0.001
OSCE_mean	OSCE_min_sofar	0.8386	<0.001
OSCE_mean	OSCE_max_sofar	0.8257	<0.001
Written_prev	Written_min_sofar	0.7698	<0.001
OSCE_prev	Written-OSCE	−0.7168	<0.001

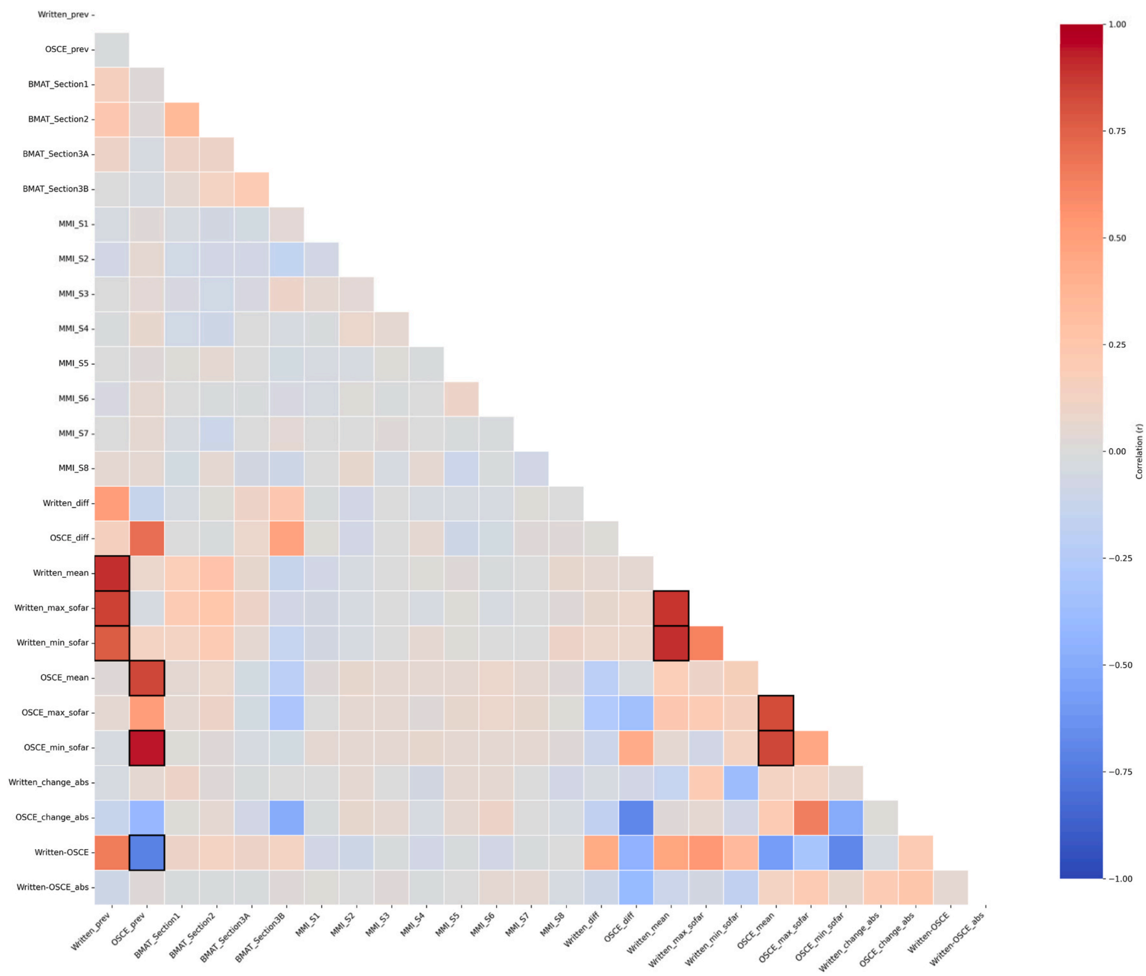

Fig. A1. Spearman correlation matrix of variables. Variables with $|\rho| \geq 0.7$ were highlighted.

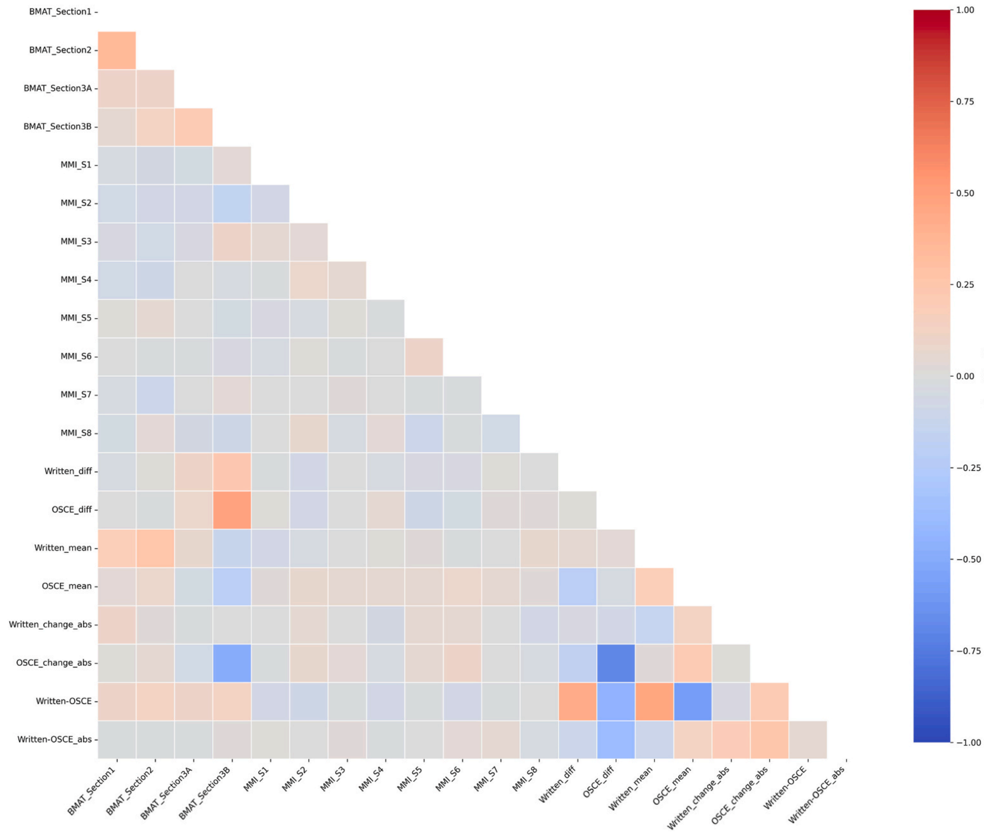

Fig. A2. Spearman correlation matrix of remaining variables. None of the remaining variables pairs exceeded the $|\rho| \geq 0.7$ threshold.

Table A2
Summary of variables and their description

Variable	Category	Description
Gender	Sociodemographic	Student's gender
Age Group	Sociodemographic	High-age group vs. standard entry-age group
Housing Type	Sociodemographic	Type of housing
Biology Background	Pre-admission	Whether the student has a good biology background
Aptitude skill (BMAT_S1)	Pre-admission	Assesses analytical reasoning and problem-solving ability
Scientific knowledge (BMAT_S2)	Pre-admission	Assesses application of scientific knowledge
Writing content quality (BMAT_S3A)	Pre-admission	Assesses clarity of argumentation and written communication
Written English quality (BMAT_S3B)	Pre-admission	Assesses language accuracy and academic writing skill
Professional role awareness (MMI_S1)	Pre-admission	Assesses awareness of the physician's professional role
Communication skills (MMI_S2)	Pre-admission	Assesses interpersonal and patient-centered communication
Leadership & teamwork (MMI_S3)	Pre-admission	Assesses collaboration, leadership, and shared decision-making
Time management (MMI_S4)	Pre-admission	Assesses workload planning, organization, and reliability
Scientific interest (MMI_S5)	Pre-admission	Assesses intellectual curiosity and interest in science
Professional integrity (MMI_S6)	Pre-admission	Assesses integrity, honesty, compassion, and altruism
Resilience & adaptability (MMI_S7)	Pre-admission	Assesses adaptability and coping under stress
Commitment & motivation (MMI_S8)	Pre-admission	Assesses motivation and commitment to medicine
Academic performance (Written_mean)	In-school	Assesses overall performance on written examinations
Clinical performance (OSCE_mean)	In-school	Assesses overall performance on clinical examinations
Academic progression (Written_diff)	In-school	Assesses progression in academic performance over time: $\text{Year}_t - \text{Year}_{t-1}$
Clinical progression (OSCE_diff)	In-school	Assesses development of clinical competence across time: $\text{Year}_t - \text{Year}_{t-1}$
Academic fluctuation (Written_change_abs)	In-school	Assesses fluctuation in academic performance: $\lvert$Written_Diff$\rvert$
Clinical fluctuation (OSCE_change_abs)	In-school	Assesses fluctuation in performance on clinical examinations: $\lvert$OSCE_Diff$\rvert$
Relative knowledge–practice (Written-OSCE)	In-school	Assesses difference of written and clinical examinations (positive = written stronger): Written score - OSCE score
Knowledge–practice gap (Written-OSCE_abs)	In-school	Assesses alignment between performance on written and clinical examinations (higher = greater imbalance): $\lvert$Written − OSCE$\rvert$

Appendix B

Table B1 shows the long-format dataset used in TD-Cox analysis. At Year 1, 1,039 students were included, of whom 113 (10.88%) experienced professionalism issues. By Year 2, the cohort comprised 842 students, with 92 events (10.93%). In Year 3, 650 students were included, and 119 events (18.31%) were observed, representing the highest proportion across the study period. The sample size continued to decrease in later years, with 455 students and 57 events (12.53%) in Year 4, and 292 students with 37 events (12.67%) in Year 5.

Table B1
Long-format dataset used in TD-Cox analysis.

Year_t	Number	Event = 0	Event = 1	Event = 0 (%)	Event = 1 (%)
1	1039	926	113	89.12	10.88
2	842	750	92	89.07	10.93
3	650	531	119	81.69	18.31
4	455	398	57	87.47	12.53
5	292	255	37	87.33	12.67

Note. Event = 1 indicates the occurrence of a professionalism issue; Event = 0 indicates no issue.

Table B2 presents the results of the multivariable TD-Cox regression model fitted on the training set, including HR, 95% CI for HR, and *p* values.

Table B2
Multivariate TD-Cox regression coefficients and hazard ratios

Variable	*p*	95% CI for HR	HR
Writing content quality (BMAT_S3A)	0.0110	[0.8249, 0.9754]	0.8970
Professional integrity (MMI_S6)	0.0189	[1.0163, 1.1969]	1.1030
Academic performance (Written_mean)	0.0571	[0.8157, 1.0030]	0.9045
Relative knowledge–practice (Written-OSCE)	0.1005	[0.7574, 1.0248]	0.8810
Communication skills (MMI_S2)	0.1224	[0.9827, 1.1587]	1.0671
Professional role awareness (MMI_S1)	0.2151	[0.9705, 1.1421]	1.0528
Written English quality (BMAT_S3B)	0.2205	[0.9688, 1.1474]	1.0543
Leadership & teamwork (MMI_S3)	0.2312	[0.9687, 1.1406]	1.0512
Knowledge-practice gap (Written-OSCE_abs)	0.2517	[0.9507, 1.2128]	1.0738
Age Group_1	0.3374	[0.9146, 1.2975]	1.0893
Biology Background_1	0.3890	[0.9061, 1.2881]	1.0804
Gender_1	0.3988	[0.7776, 1.1053]	0.9271
Academic progression (Written_diff)	0.4250	[0.8353, 1.0788]	0.9493
Clinical fluctuation (OSCE_change_abs)	0.4656	[0.7679, 1.1284]	0.9309
Clinical performance (OSCE_mean)	0.4892	[0.9159, 1.2015]	1.0491
Academic fluctuation (Written_change_abs)	0.4940	[0.8477, 1.0830]	0.9582
Scientific interest (MMI_S5)	0.5154	[0.9470, 1.1147]	1.0274
Resilience & adaptability (MMI_S7)	0.5585	[0.9004, 1.0583]	0.9762
Commitment & motivation (MMI_S8)	0.6424	[0.9395, 1.1064]	1.0196
Scientific knowledge (BMAT_S2)	0.6513	[0.9007, 1.0675]	0.9806
Aptitude skill (BMAT_S1)	0.7329	[0.9326, 1.1042]	1.0148
Time management (MMI_S4)	0.8139	[0.9311, 1.0952]	1.0098
Housing_1	0.8756	[0.8560, 1.2002]	1.0136
Clinical progression (OSCE_diff)	0.9113	[0.8276, 1.2362]	1.0115

Note. HR = hazard ratio; 95% CI for HR indicates the lower and upper bounds of the 95% confidence interval for the hazard ratio.

Appendix C

In the Kaplan–Meier estimator, for each variable, students were stratified into high- and low-score groups. For Written English quality (BMAT_S3B), in which most students achieved the maximum score (5), the *High* group was defined as those with a score of 5, and the *Low* group as those with scores below 5. For all other continuous variables, the median value served as the cut-off: scores below the median were classified as *Low*, and those at or above the median as *High*. We further examined group differences by comparing the top and bottom 10% of students on significant predictors. Figure C1 and Figure C2 present the Kaplan–Meier survival curves for subgroups stratified by pre-admission scores and in-school performance, respectively.

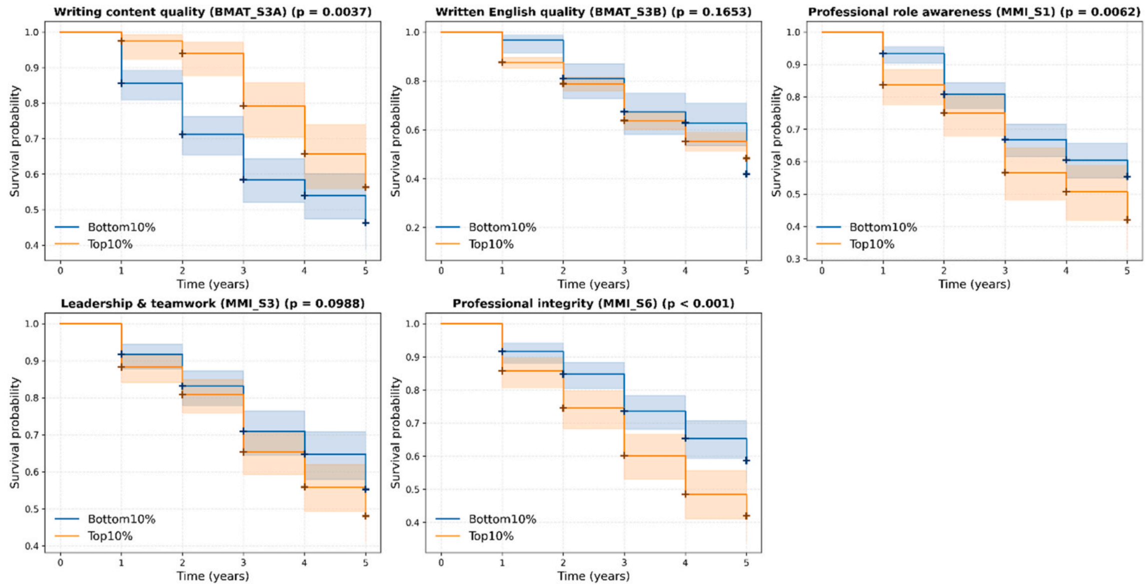

Fig. C1. Kaplan–Meier curves comparing top 10% and bottom 10% groups in the pre-admission scores. The y-axis represents the probability of remaining professional (survival probability), where higher values indicate a lower risk of professionalism issues. Flatter curves reflect a slower progression toward professionalism issues and longer survival, whereas steeper declines indicate faster lapses. Cross marks (+) indicate censored data (students who graduated or left the study without a recorded professionalism issue).

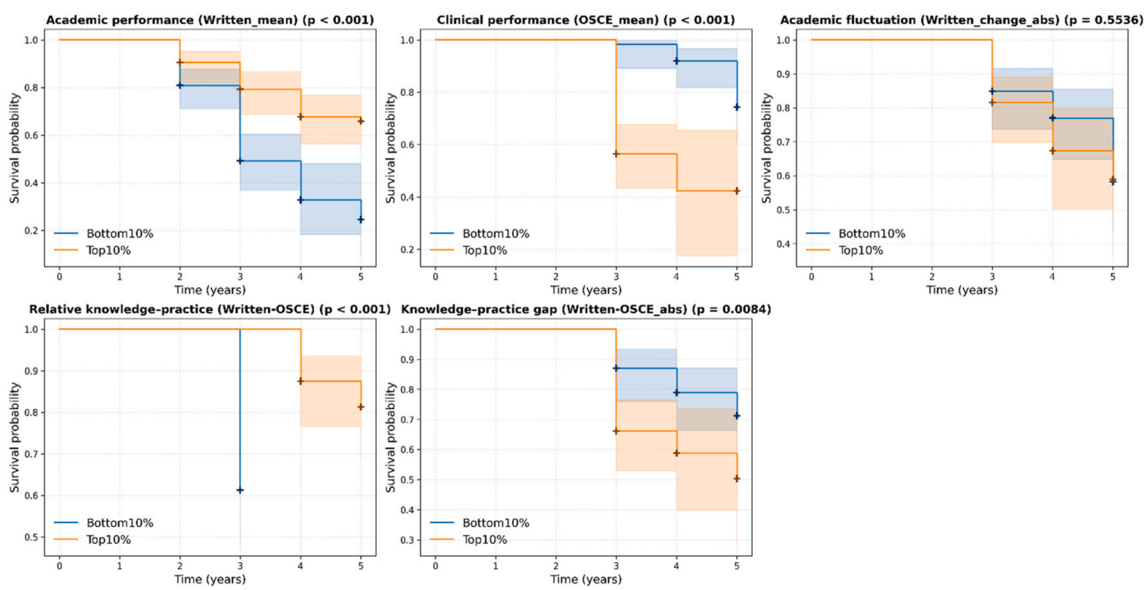

Fig. C2. Kaplan–Meier curves comparing top 10% and bottom 10% groups in the in-school performance. The y-axis represents the probability of remaining professional (survival probability), where higher values indicate a lower risk of professionalism issues. Flatter curves reflect a slower progression toward professionalism issues and longer survival, whereas steeper declines indicate faster lapses. Cross marks (+) indicate censored data (students who graduated or left the study without a recorded professionalism issue).

Appendix D

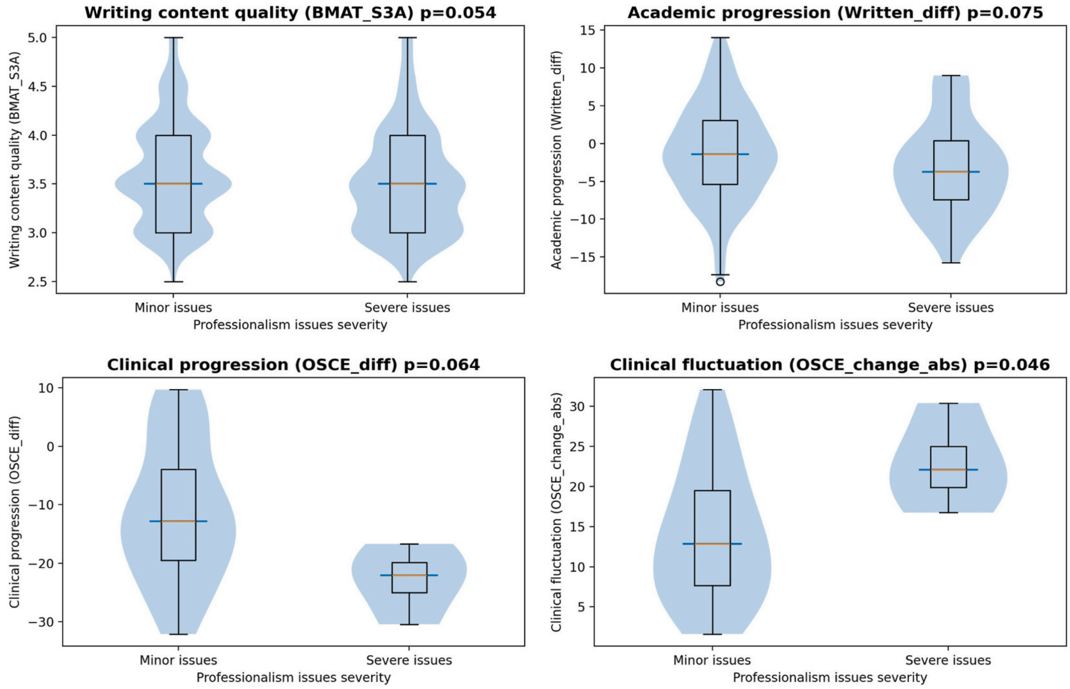

Fig. D1. Exploratory analysis of minor versus severe professionalism issues. Violin and box plots show distributional differences in predictors. Violin widths represent data density, while boxplots indicate the median and interquartile range.

Table D1 shows the statistical results for students with minor versus severe professional issues.

Table D1
Group comparison results (mino*r issues vs severe issues)*

Variable	Type	Test	*p*
Biology Background	Categorical	Chi-square	0.3228
Gender	Categorical	Chi-square	0.2424
Age Group	Categorical	Chi-square	0.1227
Housing	Categorical	Chi-square	0.9466
Aptitude skill (BMAT_S1)	Numeric	Mann–Whitney U	0.4409
Scientific knowledge (BMAT_S2)	Numeric	Mann–Whitney U	0.1806
Writing content quality (BMAT_S3A)	Numeric	Mann–Whitney U	0.0540
Written English quality (BMAT_S3B)	Numeric	Mann–Whitney U	0.3399
Professional role awareness (MMI_S1)	Numeric	Mann–Whitney U	0.7888
Leadership & teamwork (MMI_S3)	Numeric	Mann–Whitney U	0.5888
Professional integrity (MMI_S6)	Numeric	Mann–Whitney U	0.1963
Communication skills (MMI_S2)	Numeric	Mann–Whitney U	0.2123
Time management (MMI_S4)	Numeric	Mann–Whitney U	0.7704
Scientific interest (MMI_S5)	Numeric	Mann–Whitney U	0.3205
Resilience & adaptability (MMI_S7)	Numeric	Mann–Whitney U	0.2080
Commitment & motivation (MMI_S8)	Numeric	Mann–Whitney U	0.6278
Academic progression (Written_diff)	Numeric	*t*-test	0.0752
Clinical progression (OSCE_diff)	Numeric	*t*-test	0.0639
Academic performance (Written_mean)	Numeric	Mann–Whitney U	0.8108
Clinical performance (OSCE_mean)	Numeric	Mann–Whitney U	0.5393
Academic fluctuation (Written_change_abs)	Numeric	Mann–Whitney U	0.3931
Clinical fluctuation (OSCE_change_abs)	Numeric	*t*-test	0.0463
Relative knowledge-practice (Written-OSCE)	Numeric	Mann–Whitney U	0.2315
Knowledge-practice gap (Written-OSCE_abs)	Numeric	Mann–Whitney U	0.4015

Appendix E

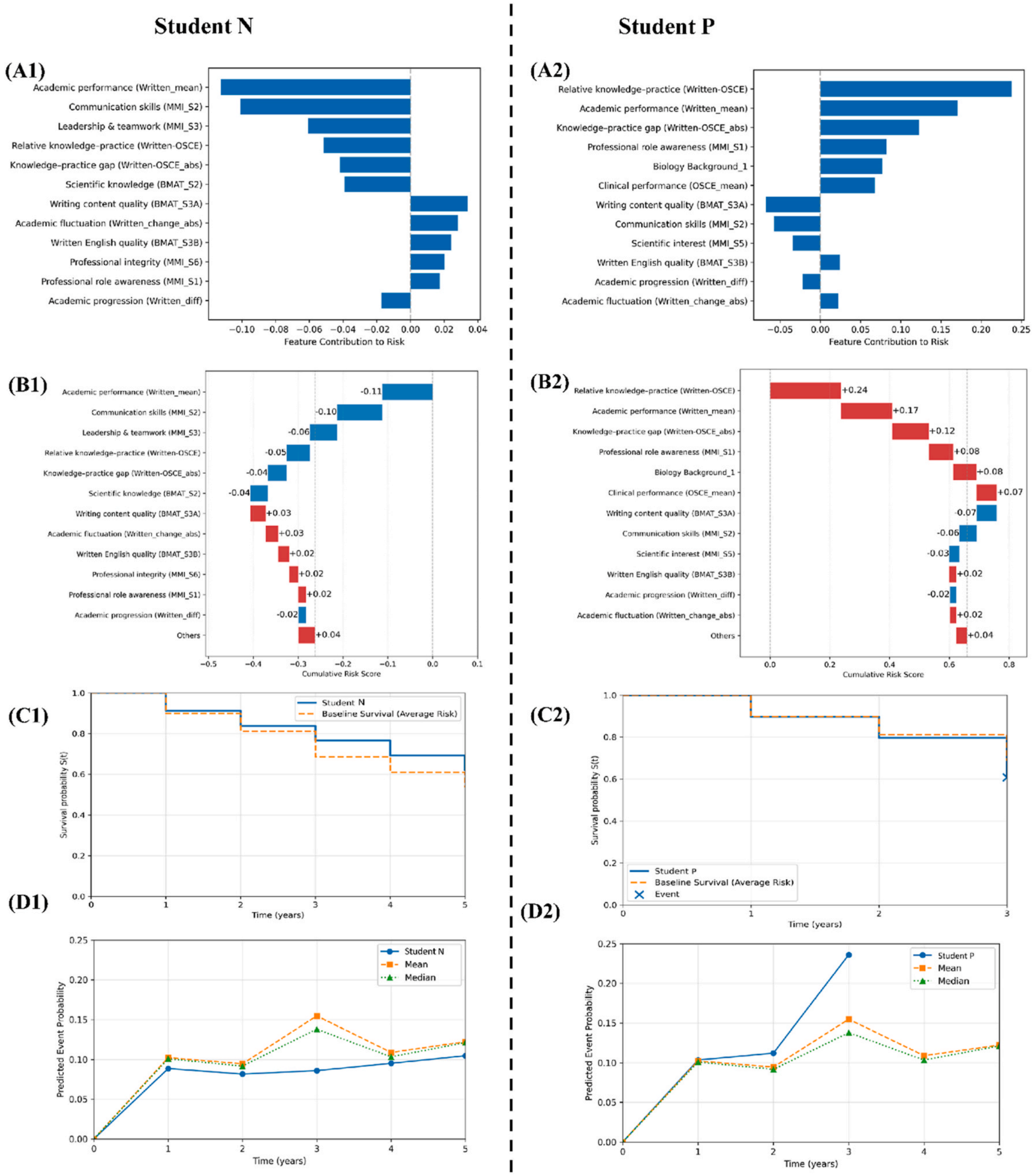

Fig. E1. *Individual-level risk analysis results for two representative students: Student N (left panels), who did not experience professionalism issues, and Student P (right panels), who developed an issue in Year 3. Panels A1–A2 show the most influential predictors for each student based on individualized feature contributions. Positive values indicate risk-enhancing effects; negative values indicate protective effects. Panels B1–B2 show the cumulative effects of these predictors using waterfall visualizations. Red bars show risk-enhancing effects, blue bars protective effects. Panels C1–C2 present individualized survival curves, compared against the cohort baseline survival. The y-axis indicates survival probability S(t). Panels D1–D2 display the predicted probability trajectories of experiencing a professionalism issue over time, alongside cohort mean and median reference curves. The y-axis represents event probability, where higher values indicate a greater likelihood of professionalism issues at a given time point.*

Appendix F

Sensitivity Analysis 1: Predicting Professionalism Issues without BMAT Sections

As BMAT was not used by all medical schools as a pre-admission assessment, we have also repeated our analysis without BMAT variables. Table F1 summarizes the predictive performance after removing the BMAT sections, broadly aligning with the main results in Table 4. Table F2 and Figure F1 show the variables with the predictive power for predicting professionalism issues. Professional integrity (MMI_S6), Professional role awareness (MMI_S1), and Academic performance (Written_mean) were statistically significant ($p < 0.05$), with hazard ratios of 1.2130, 1.1339, and 0.8413, respectively. This is consistent with our current findings.

Table F1
Performance of TD-Cox model on the test dataset (remove BMAT sections)

Year	C-index	Brier	Accuracy	F1	Sample Size
1	0.6738	0.0893	0.8702	0.8584	208
2	0.6142	0.1280	0.7574	0.7344	169
3	0.8130	0.1565	0.7846	0.7846	130
4	0.4359	0.1334	0.7826	0.7570	92
5	0.5714	0.0975	0.7969	0.7899	64

Table F2
Multivariate TD-Cox regression coefficients and hazard ratios (remove BMAT sections)

Variable	p	95% CI for HR	HR
Professional integrity (MMI_S6)	**< 0.001**	**[1.0987, 1.3392]**	1.2130
Relative knowledge-practice (Written-OSCE)	0.1030	[0.6388, 1.0420]	0.8159
Academic performance (Written_mean)	**0.0250**	**[0.7234, 0.9785]**	0.8413
Knowledg-Practice gap (Written-OSCE_abs)	0.0582	[0.9952, 1.3246]	1.1482
Professional role awareness (MMI_S1)	**0.0110**	**[1.0292, 1.2492]**	1.1339
Academic fluctuation (Written_change_abs)	0.1318	[0.7726, 1.0343]	0.8939
Leadership & teamwork (MMI_S3)	0.0977	[0.9850, 1.1975]	1.0861
Clinical performance (OSCE_mean)	0.4954	[0.8725, 1.3256]	1.0755

Note. HR = hazard ratio; CI = confidence interval; Statistical significance is defined as $p < 0.05$, which corresponds to 95% confidence intervals for HR that do not include 1.

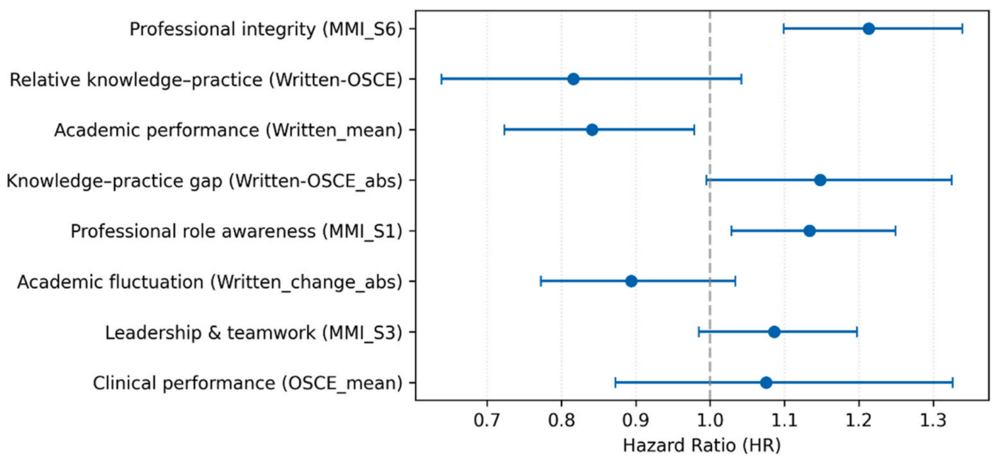

Fig. F1. Forest plot displays HR with 95% CI from the TD-Cox regression analysis (remove BMAT sections). An HR greater than 1 indicates increased risk; an HR less than 1 indicates decreased risk, while HR estimates crossing 1 suggest non-significant effects.

Descriptive Statistics for MMI Dimensions by Professionalism Issue Status

We conducted descriptive analyses for two MMI dimensions (MMI_S1 (Professional Role Awareness) and MMI_S6 (Professional Integrity)) across academic years from 2016 to 2023. As there were only four records in 2016, these were combined with 2017 data for analysis (hereafter referred to as 2017). For each dimension, we calculated the mean, standard deviation (SD), median, and first (Q1) and third (Q3) quartiles, stratified by students with and without professionalism issues.

For MMI_S1, in most years (2017, 2018, 2019, 2020, 2022, and 2023), students in the issue group exhibited higher mean scores compared to those without issues. Across all years, the issue group consistently demonstrated equal or higher medians and quartiles (median, Q1, Q3) relative to the no-issue group (see Table F3 and Figure F2). Similarly, for MMI_S6, students in the issue group showed higher mean scores in most years (2018, 2019, 2020, 2021, 2022, and 2023). For every year, the issue group had equal or higher medians and quartiles compared to students without issues (see Table F4 and Figure F3).

These descriptive findings align with our previous analyses, indicating that higher MMI_S1 and MMI_S6 scores are associated with an increased risk of professionalism issues.

Table F3
Descriptive statistics of Professional role awareness (MMI_S1) scores between students with and without professionalism issues.

Year	No_Issue Mean (SD)	Issue_Mean (SD)	No_Issue Median	Issue Median	No_Issue Q1	Issue Q1	No_Issue Q3	Issue Q3
2017	3.6190 (0.8702)	3.7500 (0.7746)	4	4	3	3	4	4
2018	3.6705 (0.8266)	3.8367 (0.7173)	4	4	3	3	4	4
2019	3.5968 (0.7566)	3.7317 (0.7706)	4	4	3	3	4	4
2020	3.6038 (0.7261)	3.6792 (0.7009)	4	4	3	3	4	4
2021	3.8684 (0.7890)	3.8000 (0.8767)	4	4	3	3	4	4

(*continued on next page*)

Table F3 (*continued*)

Year	No_Issue Mean (SD)	Issue_Mean (SD)	No_Issue Median	Issue Median	No_Issue Q1	Issue Q1	No_Issue Q3	Issue Q3
2022	3.6800 (0.9415)	3.7188 (0.8992)	4	4	3	3	4	4
2023	3.9405 (0.7339)	4.0938 (0.7285)	4	4	3	4	4	5

Note: "No_Issue" refers to students without professionalism issues; "Issue" refers to students with professionalism issues. SD = standard deviation; Q1 = first quartile; Q3 = third quartile.

Table F4
Descriptive Statistics of Professional integrity scores (MMI_S6) between students with and without professionalism issues.

Year	No_Issue Mean (SD)	Issue_Mean (SD)	No_Issue Median	Issue Median	No_Issue Q1	Issue Q1	No_Issue Q3	Issue Q3
2017	3.7619 (0.8381)	3.500 (0.6325)	4	4	3	3	4	4
2018	3.5909 (0.7825)	3.7551 (0.9021)	3.5	4	3	3	4	4
2019	3.8548 (0.7208)	4.0000 (0.7370)	4	4	3	4	4	4.75
2020	3.8491 (0.8257)	4.0943 (0.8149)	4	4	3	4	4	5
2021	3.8026 (0.8330)	4.0333 (0.7710)	4	4	3	4	4	5
2022	3.8300 (0.8883)	3.9375 (0.7741)	4	4	3	3	4	4
2023	3.9405 (0.8118)	4.0156 (0.7866)	4	4	3	4	5	5

Note: "No_Issue" refers to students without professionalism issues; "Issue" refers to students with professionalism issues. SD = standard deviation; Q1 = first quartile; Q3 = third quartile.

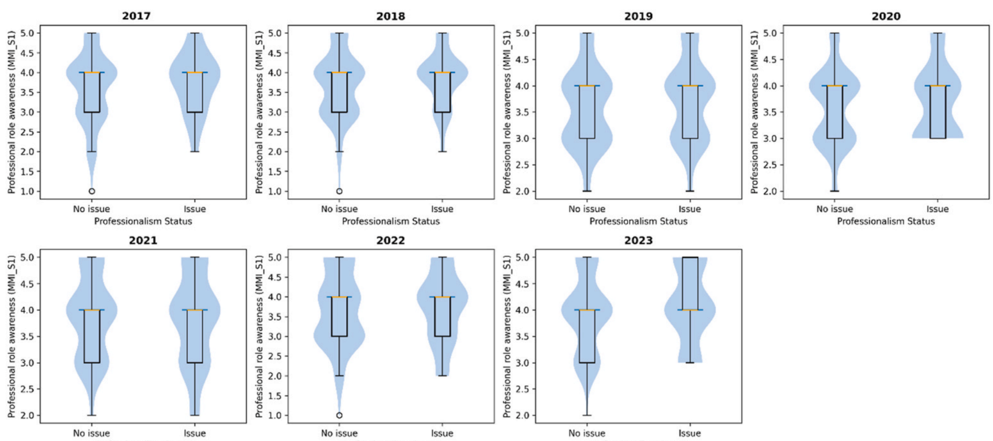

Fig. F2. Annual comparison of Professional role awareness (MMI_S1) scores between students with and without professionalism issues. Violin and box plots show distributional differences in the two groups. Violin widths represent data density, while boxplots indicate the median and interquartile range.

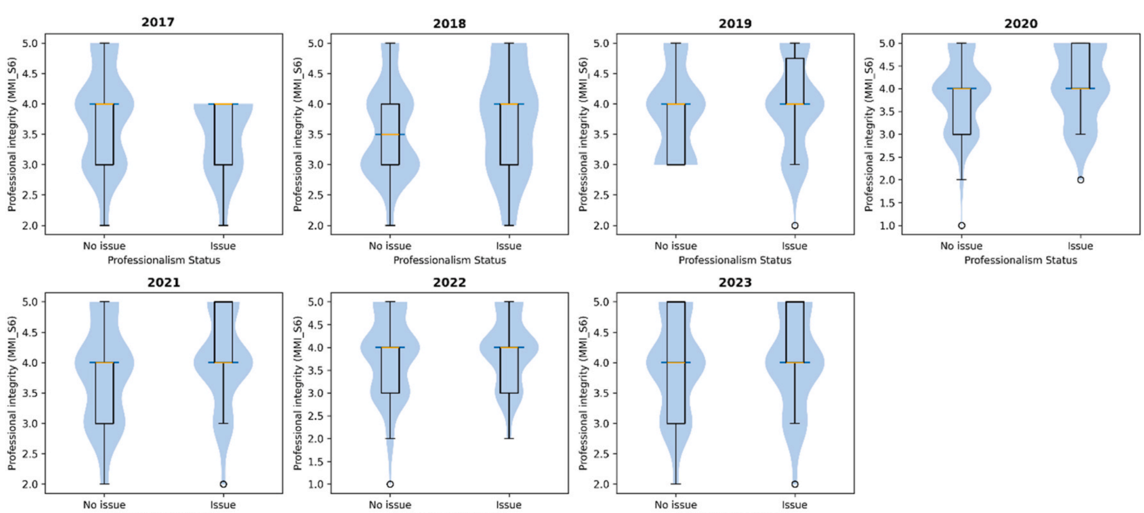

Fig. F3. Annual comparison of Professional integrity (MMI_S6) scores between students with and without professionalism issues. Violin and box plots show distributional differences in the two groups. Violin widths represent data density, while boxplots indicate the median and interquartile range.

Sensitivity Analysis 2: Predicting Professionalism Issues after Excluding a Specific CHERI Comment

To account for potential bias in the identification of professionalism issues derived from locally defined CHERI comments, we conducted a sensitivity analysis. Specifically, we removed one CHERI comment: "Failure to submit Course Feedback and/or Evaluation Form(s)" (n = 54). Professionalism issues were then redefined. The longitudinal dataset was reconstructed, and the TD-Cox model was re-trained following the same procedure as in the main analysis.

The results were largely consistent with the main findings. Table F5 shows performance patterns consistent with the main results reported in Table 4. Key predictors were also consistent with the main findings (Table F6). In particular, higher Writing content quality (BMAT_S3A) and Academic performance (Written_mean) were associated with lower risk, whereas higher Professional role awareness (MMI_S1), Professional integrity (MMI_S6), and Written English quality (BMAT_S3B) scores were associated with increased risk.

Table F5
Performance of TD-Cox model on the test dataset (remove one CHERI Comment)

Year	C-index	Brier	Accuracy	F1	Sample Size
1	0.6243	0.0703	0.8750	0.8750	208
2	0.6467	0.1084	0.7989	0.7921	174
3	0.7644	0.1543	0.7737	0.7779	137
4	0.3960	0.1333	0.7742	0.7531	93
5	0.3678	0.0868	0.8438	0.8438	64

Table F6
Multivariate TD-Cox regression coefficients and hazard ratios (*remove one CHERI Comment*)

Variable	*p*	95% CI for HR	HR
Written English quality (BMAT_S3B)	**0.0012**	**[1.0859, 1.3965]**	1.2314
Academic performance (Written_mean)	**0.0061**	**[0.6690, 0.9353]**	0.7910
Professional integrity (MMI_S6)	**0.0019**	**[1.0641, 1.3155]**	1.1831
Clinical performance (OSCE_mean)	0.1876	[0.9299, 1.4495]	1.1610
Writing content quality (BMAT_S3A)	**0.0079**	**[0.7743, 0.9621]**	0.8631
Academic fluctuation (Written_change_abs)	0.0712	[0.7520, 1.0119]	0.8723
Professional role awareness (MMI_S1)	**0.0439**	**[1.0029, 1.2333]**	1.1122
Relative knowledge-practice (Written–OSCE)	0.4480	[0.6945, 1.1747]	0.9033
Knowledge-Practice gap (Written–OSCE_abs)	0.2351	[0.9431, 1.2697]	1.0942
Leadership & teamwork (MMI_S3)	0.1789	[0.9676, 1.1931]	1.0744

Note. HR = hazard ratio; CI = confidence interval; Statistical significance is defined as $p < 0.05$, which corresponds to 95% confidence intervals for HR that do not include 1.

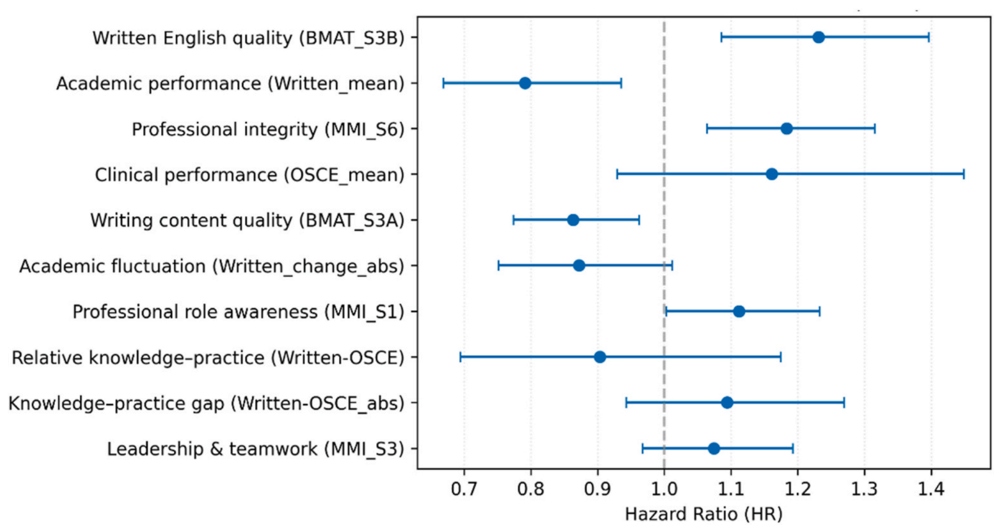

Fig. F4. Forest plot displays HR with 95% CI from the TD-Cox regression analysis (remove one CHERI Comment). An HR greater than 1 indicates increased risk; an HR less than 1 indicates decreased risk, while HR estimates crossing 1 suggest non-significant effects.

Robustness Check

To evaluate the robustness of the TD-Cox model, we conducted robustness check. The entire modeling was repeated using 10 different random seeds. For each seed, the model was independently trained and evaluated, and performance metrics, including C-index, Brier score, Accuracy, and F1-score, were computed on the test set. The final results are reported as the mean and standard deviation (mean ± std) across the 10 runs.

Table F7 shows that performance across academic years broadly aligns with the main results in Table 4. The model achieved higher C-index values in Year 1 and Year 3, and lower performance in Year 4 and Year 5. Consistent with Table 4, the Brier score remained low, while Accuracy and F1 were relatively high (>0.75) across all five years.

Table F7
Robust check results: performance of TD-Cox model on the test dataset

Year	C-index Mean (Std)	Brier Mean (Std)	Accuracy Mean (Std)	F1 Mean (Std)
1	0.6936 (0.0712)	0.0920 (0.0161)	0.8385 (0.0198)	0.8399 (0.0250)
2	0.6029 (0.0677)	0.1015 (0.0153)	0.8268 (0.0259)	0.8212 (0.0274)
3	0.7558 (0.0404)	0.1394 (0.0211)	0.7921 (0.0315)	0.7869 (0.0362)
4	0.4691 (0.0589)	0.1136 (0.0204)	0.7532 (0.0326)	0.7559 (0.0265)
5	0.4840 (0.0646)	0.1306 (0.0260)	0.7661 (0.0305)	0.7512 (0.0395)

Data availability

The data can be obtained by sending request e-mails to the corresponding author.

References

Ainsworth, M. A., & Szauter, K. M. (2006). Medical student professionalism: Are we measuring the right behaviors? A comparison of professional lapses by students and physicians. *Academic Medicine, 81*(10), S83–S86.

Ali, S., Sadiq Hashmi, M. S., Umair, M., Beg, M. A., & Huda, N. (2019). Multiple mini-interviews: Current perspectives on utility and limitations. *Advances in medical education and practice*.

Altman, D., De Stavola, B., Love, S., & Stepniewska, K. (1995). Review of survival analyses published in cancer journals. *British Journal of Cancer, 72*(2), 511–518.

Ameri, S., Fard, M. J., Chinnam, R. B., & Reddy, C. K. (2016). Survival analysis based framework for early prediction of student dropouts. *Proceedings of the 25th ACM international on conference on information and knowledge management*.

Andersen, P. K. (1991). Survival analysis 1982–1991: The second decade of the proportional hazards regression model. *Statistics in Medicine, 10*(12), 1931–1941.

Andersen, P. K., & Gill, R. D. (1982). Cox's regression model for counting processes: A large sample study. *Annals of Statistics*, 1100–1120.

Bland, J. M., & Altman, D. G. (2004). The logrank test. *BMJ, 328*(7447), 1073.

Brennan, M. D., & Monson, V. (2014). Professionalism: Good for patients and health care organizations. *Mayo Clinic Proceedings, 89*(5), 644–652.

Bruinsma, M., & Jansen, E. P. (2009). When will i succeed in my first-year diploma? Survival analysis in dutch higher education. *Higher Education Research and Development, 28*(1), 99–114.

Cambridge Assessment Admissions Testing. (2003). BioMedical admissions Test (BMAT). https://www.admissionstesting.org/for-test-takers/bmat/.

Coates, W. C. (2025). Precision education–a call to action to transform medical education. *International Journal of Emergency Medicine, 18*(1), 21.

Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B, 34*(2), 187–202.

Cruess, R. L., & Cruess, S. R. (2008). Expectations and obligations: Professionalism and medicine's social contract with society. *Perspectives in Biology and Medicine, 51*(4), 579–598.

Cruess, R. L., Cruess, S. R., Boudreau, J. D., Snell, L., & Steinert, Y. (2014). Reframing medical education to support professional identity formation. *Academic Medicine, 89*(11), 1446–1451.

Cruess, R. L., Cruess, S. R., & Johnston, S. E. (2000). Professionalism: An ideal to be sustained. *The Lancet, 356*(9224), 156–159.

Cuddy, M. M., Young, A., Gelman, A., Swanson, D. B., Johnson, D. A., Dillon, G. F., & Clauser, B. E. (2017). Exploring the relationships between usmle performance and disciplinary action in practice: A validity study of score inferences from a licensure examination. *Academic Medicine, 92*(12), 1780–1785.

da Rosa, M. I., Ceretta, L. B., Martins, M. A., Colonetti, L., Colonetti, T., Grande, A. J., & Tempski, P. (2024). The effects of academic unprofessional behaviour on disciplinary action by medical boards: Systematic review and meta-analysis. *The Clinical Teacher, 21*, (4), Article e13740.

Donders, A. R. T., Van Der Heijden, G. J., Stijnen, T., & Moons, K. G. (2006). A gentle introduction to imputation of missing values. *Journal of Clinical Epidemiology, 59*(10), 1087–1091.

Du Prel, J.-B., Röhrig, B., Hommel, G., & Blettner, M. (2010). Choosing statistical tests: Part 12 of a series on evaluation of scientific publications. *Deutsches Ärzteblatt International, 107*(19), 343.

Eva, K. W., Rosenfeld, J., Reiter, H. I., & Norman, G. R. (2004). An admissions osce: The multiple mini-interview. *Medical Education, 38*(3), 314–326.

Foo, J., Ilic, D., Rivers, G., Evans, D. J., Walsh, K., Haines, T. P., Paynter, S., Morgan, P., & Maloney, S. (2018). Using cost-analyses to inform health professions education–the economic cost of pre-clinical failure. *Medical Teacher, 40*(12), 1221–1230.

George, B., Seals, S., & Aban, I. (2014). Survival analysis and regression models. *Journal of Nuclear Cardiology, 21*(4), 686–694.

Glenn, W. B. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review, 78*(1), 1–3.

Gury, N. (2011). Dropping out of higher education in France: A micro-economic approach using survival analysis. *Education Economics, 19*(1), 51–64.

Harden, R. M., Stevenson, M., Downie, W. W., & Wilson, G. (1975). Assessment of clinical competence using objective structured examination. *Br Med J, 1*(5955), 447–451.

Harrell Jr, F. E., Lee, K. L., & Mark, D. B. (1996). Multivariable prognostic models: Issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Statistics in Medicine, 15*(4), 361–387.

Helmich, E., Yeh, H.-M., Kalet, A., & Al-Eraky, M. (2017). Becoming a doctor in different cultures: Toward a cross-cultural approach to supporting professional identity formation in medicine. *Academic Medicine, 92*(1), 58–62.

Hoffman, L. A., Shew, R. L., Vu, T. R., Brokaw, J. J., & Frankel, R. M. (2016). Is reflective ability associated with professionalism lapses during medical school? *Academic Medicine, 91*(6), 853–857.

Igarashi, R. (2025). Medical students' unprofessional behavior and educators' support. *Korean Journal of Medical Education, 37*(1), 47.

Kaplan, E. L., & Meier, P. (1958). Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association, 53*(282), 457–481.

Keshmiri, F. (2025). The experiences of unprofessionalism among students in dental education: A qualitative study. *BMC Medical Education, 25*(1), 24.

Keshmiri, F., & Raadabadi, M. (2023). Perception and engagement in unprofessional behaviors of medical students and residents: A mixed-method study. *BMC Primary Care, 24*(1), 191.

Krupat, E., Dienstag, J. L., Padrino, S. L., Mayer Jr, J. E., Shore, M. F., Young, A., Chaudhry, H. J., Pelletier, S. R., & Reis, B. Y. (2020). Do professionalism lapses in medical school predict problems in residency and clinical practice? *Academic Medicine, 95*(6), 888–895.

Labrador, M. M., Vargas, G. R. G., Alvarado, J., & Caıcedo, M. (2019). Survival and risk analysis in MOOCs. *The Turkish Online Journal of Distance Education, 20*(4), 149–159.

Mak-van der Vossen, M., Teherani, A., van Mook, W., Croiset, G., & Kusurkar, R. A. (2020). How to identify, address and report students' unprofessional behaviour in medical school. *Medical Teacher, 42*(4), 372–379.

Mak-van der Vossen, M. C., van Mook, W. N., Kors, J. M., van Wieringen, W. N., Peerdeman, S. M., Croiset, G., & Kusurkar, R. A. (2016). Distinguishing three unprofessional behavior profiles of medical students using latent class analysis. *Academic Medicine, 91*(9), 1276–1283.

Mak-Van Der Vossen, M., Van Mook, W., Van Der Burgt, S., Kors, J., Ket, J. C., Croiset, G., & Kusurkar, R. (2017). Descriptors for unprofessional behaviours of medical students: A systematic review and categorisation. *BMC Medical Education, 17*, 1–12.

Martínez-Carrascal, J. A., Hlosta, M., & Sancho-Vinuesa, T. (2023). Using survival analysis to identify populations of learners at risk of withdrawal: Conceptualization and impact of demographics. *International Review of Research in Open and Distance Learning, 24*(1), 1–21.

McLachlan, J. C., Finn, G., & Macnaughton, J. (2009). The conscientiousness index: A novel tool to explore students' professionalism. *Academic Medicine, 84*(5), 559–565.

Ministry of Education (Singapore). (2025). MOE financial assistance schemes to benefit an additional 31,000 students. https://www.moe.gov.sg/news/press-releases/20251016-moe-financial-assistance-schemes-to-benefit-an-additional-31000-students.

Monrouxe, L. V., Rees, C. E., & Hu, W. (2011). Differences in medical students' explicit discourses of professionalism: Acting, representing, becoming. *Medical Education, 45*(6), 585–602.

Ohno-Machado, L. (2001). Modeling medical prognosis: Survival analysis techniques. *Journal of Biomedical Informatics, 34*(6), 428–439.

O'sullivan, H., Van Mook, W., Fewtrell, R., & Wass, V. (2012). Integrating professionalism into the curriculum: Amee guide no. 61. *Medical Teacher, 34*(2), e64–e77.

Papadakis, M. A., Osborn, E., Cooke, M., & Healy, K. (1999). A strategy for the detection and evaluation of unprofessional behavior in medical students. university of california, sanfrancisco school of medicine clinical clerkships operation committee. *Academic Medicine, 74*(9), 980–990.

Papadakis, M. A., Teherani, A., Banach, M. A., Knettler, T. R., Rattner, S. L., Stern, D. T., Veloski, J. J., & Hodgson, C. S. (2005). Disciplinary action by medical boards and prior behavior in medical school. *New England Journal of Medicine, 353*(25), 2673–2682.

Paton, L. W., Tiffin, P. A., Smith, D., Dowell, J. S., & Mwandigha, L. M. (2018). Predictors of fitness to practise declarations in UK medical undergraduates. *BMC Medical Education, 18*(1), 68.

Peduzzi, P., Concato, J., Feinstein, A. R., & Holford, T. R. (1995). Importance of events per independent variable in proportional hazards regression analysis II. Accuracy and precision of regression estimates. *Journal of Clinical Epidemiology, 48*(12), 1503–1510.

Plank, S., DeLuca, S., & Estacion, A. (2005). *Dropping out of high school and the place of career and technical education: A survival analysis of surviving high school*. National Research Center for Career and Technical Education.

Project of The Abim Foundation, et al. (2002). Medical professionalism in the new millennium: A physician charter. *Acta Clinica Belgica, 57*(4), 169–171.

Rajasekaran, S. K. (2024). Precision education-the future of medical education. *Journal of Medical Education and Practice, 1*(1), 17–19.

Roberts, C., Burgess, A., Mossman, K., & Kumar, K. (2025). Professional judgement: A social practice perspective on a multiple mini-interview for specialty training selection. *BMC Medical Education, 25*(1), 18.

Sahota, G., Tiffin, P. A., Smith, D., Tyrrell, E., Hampshire, M., & Taggar, J. (2025). Situational judgement testing at different stages of undergraduate medical training and the risk of professionalism lapses: A cohort study. *Medical Education, 60*(3), 296–306.

Song, X., & Willy, M. J. (2024). Exploring unprofessional behaviors and biased perceptions in the clinical environment: Students' perspectives. *Medical Science Educator, 34*(5), 1049–1057.

Spearman, C. (1961). *The proof and measurement of association between two things*.

Steck, H., Krishnapuram, B., Dehing-Oberije, C., Lambin, P., & Raykar, V. C. (2007). On ranking in survival analysis: Bounds on the concordance index. *Advances in Neural Information Processing Systems, 20*.

Swick, H. M. (2000). Toward a normative definition of medical professionalism. *Academic Medicine, 75*(6), 612–616.

Triola, M. M., & Burk-Rafel, J. (2023). Precision medical education. *Academic Medicine, 98*(7), 775–781.

Wang, P., Li, Y., & Reddy, C. K. (2019). Machine learning for survival analysis: A survey. *ACM Computing Surveys, 51*(6), 1–36.

Yadav, H., Jegasothy, R., Ramakrishnappa, S., Mohanraj, J., & Senan, P. (2019). Unethical behavior and professionalism among medical students in a private medical university in Malaysia. *BMC Medical Education, 19*, 1–5.

Yates, J. (2014). "concerns" about medical students' adverse behaviour and attitude: An audit of practice at nottingham, with mapping to gmc guidance. *BMC Medical Education, 14*, 1–9.

Yates, J., & James, D. (2010). Risk factors at medical school for subsequent professional misconduct: Multicentre retrospective case-control study. *BMJ, 340*.

Zhuhadar, L., Daday, J., Marklin, S., Kessler, B., & Helbig, T. (2019). Using survival analysis to discovering pathways to success in mathematics. *Computers in Human Behavior, 92*, 487–495.

Ziring, D., Danoff, D., Grosseman, S., Langer, D., Esposito, A., Jan, M. K., Rosenzweig, S., & Novack, D. (2015). How do medical schools identify and remediate professionalism lapses in medical students? A study of us and canadian medical schools. *Academic Medicine, 90*(7), 913–920.